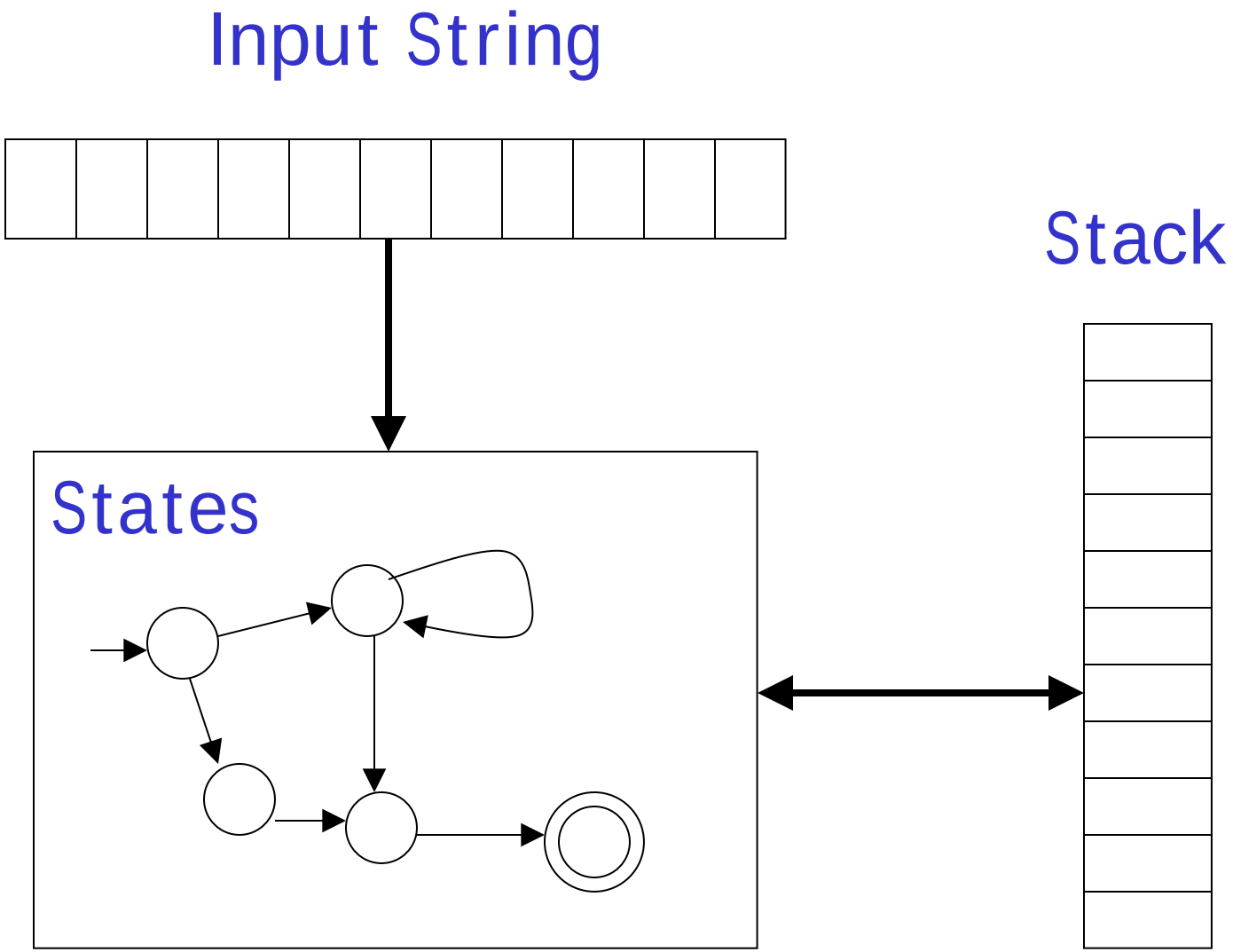
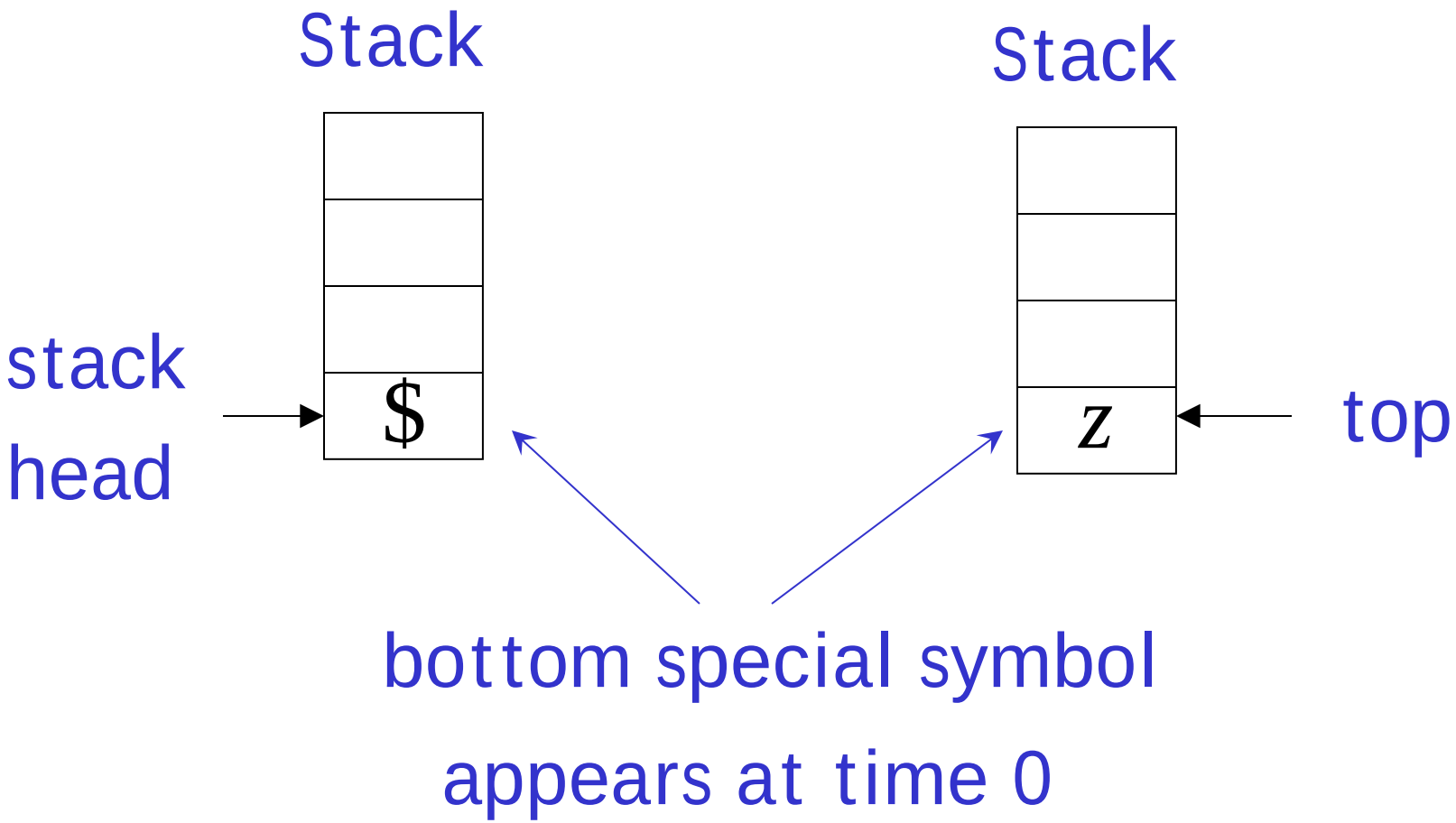
Pushdown Automata
PDAs

Original Slides by C. Busch, LSU. Modified by A. H. Özer, Marmara University.

Pushdown Automaton - PDA



Initial Stack Symbol

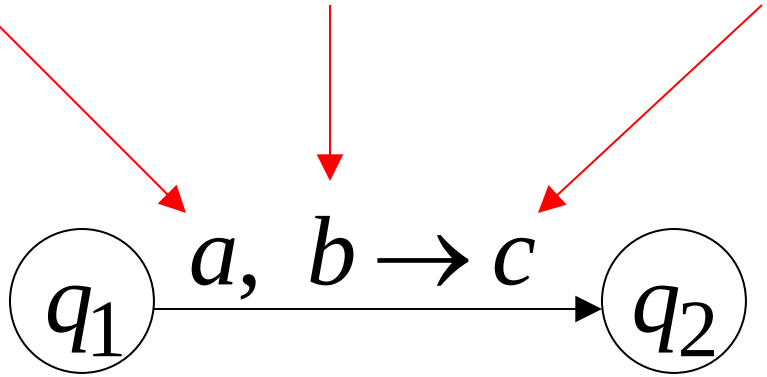


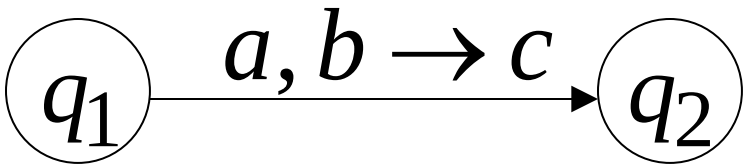
The States

Input symbol

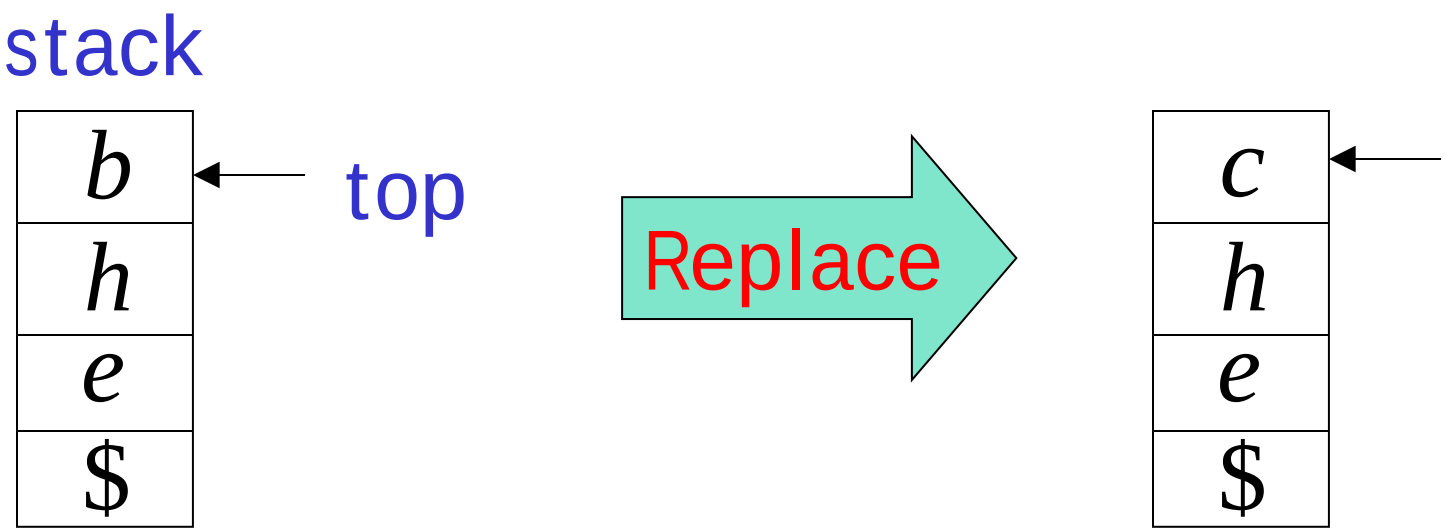
Pop symbol

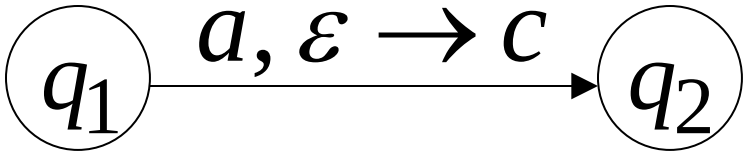
Push symbol



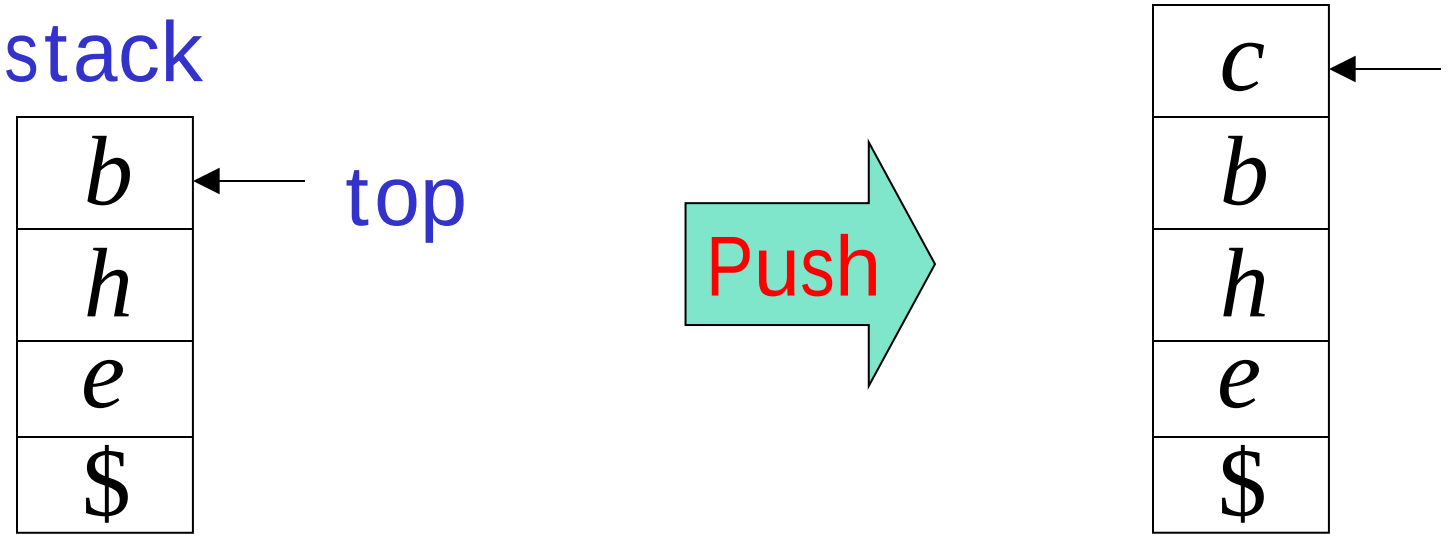

q_1
$a, b \rightarrow c$
q_2

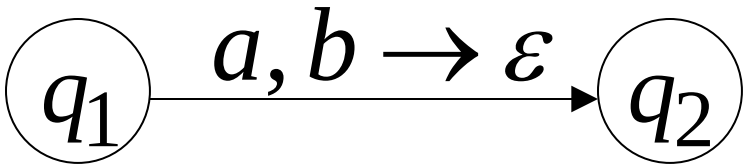

input
…
a
…
…
a
…


stack
b
h
e
$
top
Replace
c
h
e
$

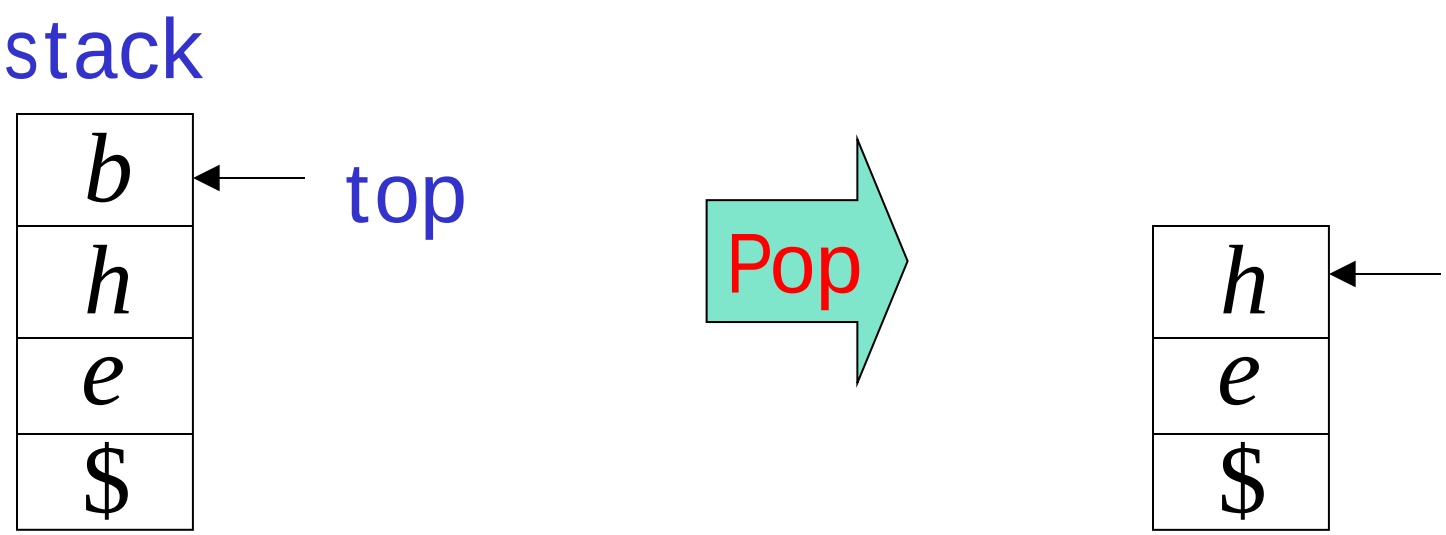

q_1
$a, \varepsilon \rightarrow c$
q_2

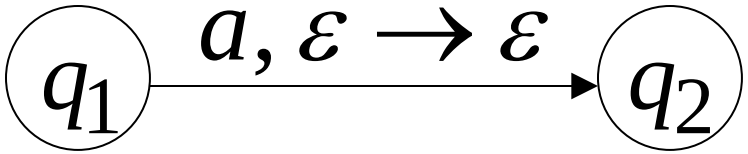

input
…
a
…
…
a
…


stack
b
top
h
e
$
Push
c
b
h
e
$

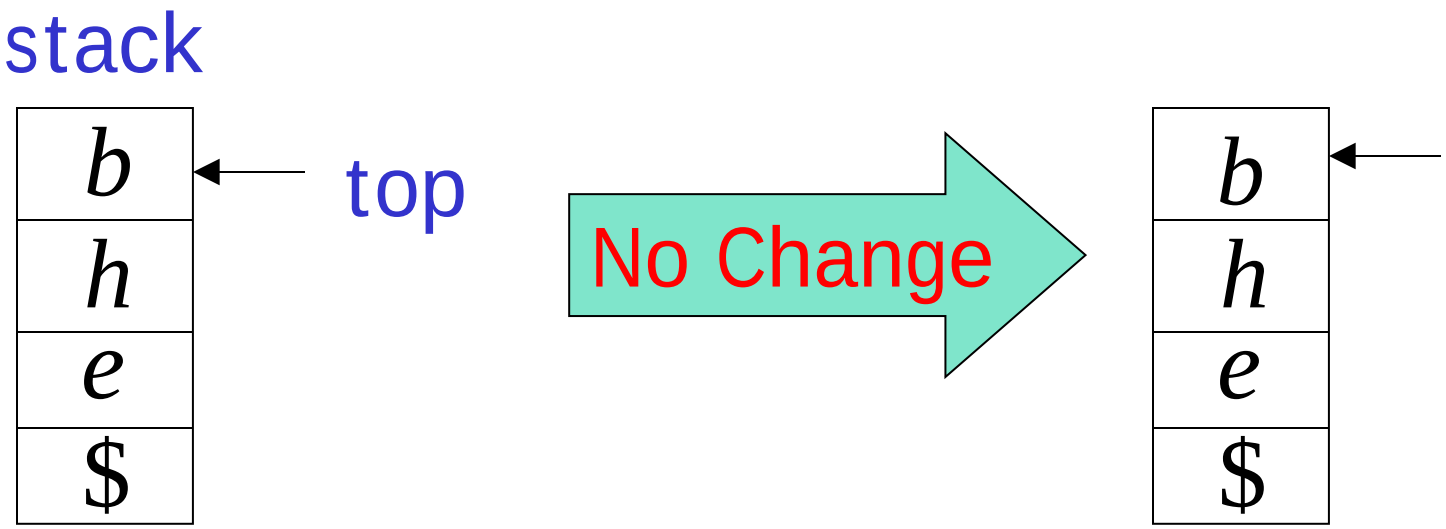

q_1
$a, b \to \varepsilon$
q_2


input
…
a
…
…
a
…


stack
b
top
h
e
$
Pop
h
e
$


q_1
$a, \varepsilon \rightarrow \varepsilon$
q_2

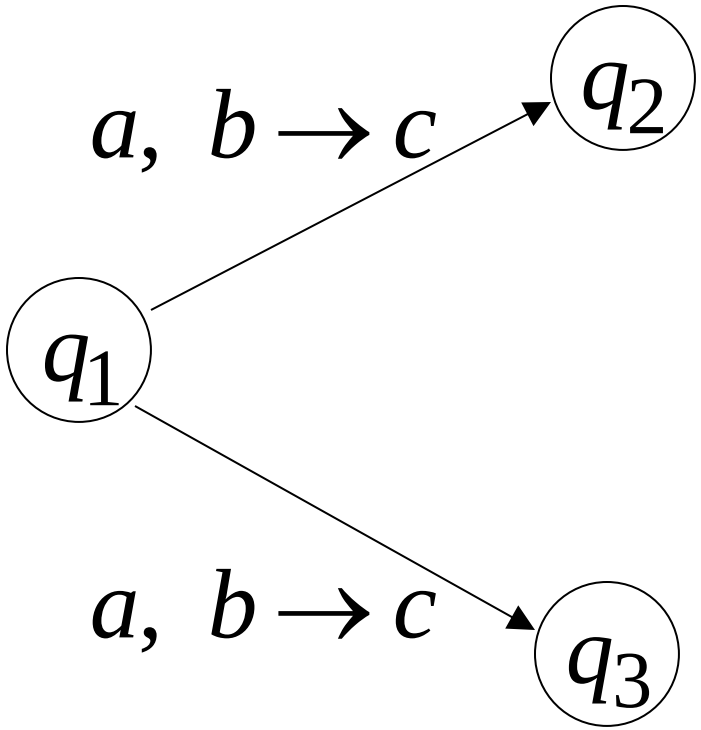

input
…
a
…
…
a
…

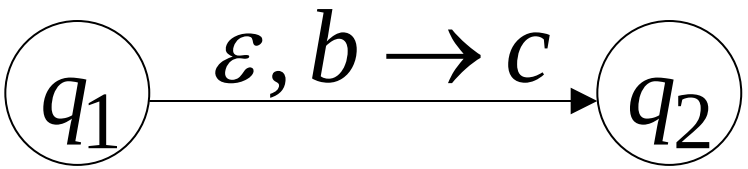

stack
b
h
e
$
top
No Change
b
h
e
$

Non-Determinism

PDAs are non-deterministic

Allowed non-deterministic transitions

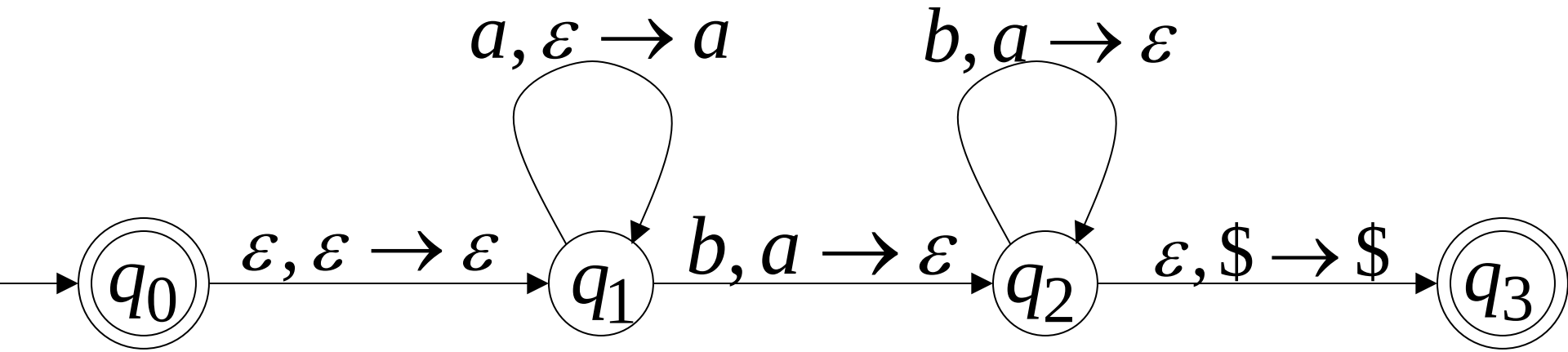




ε – transition

Example PDA

PDA M : $L(M) = \{a^n b^n : n \geq 0\}$



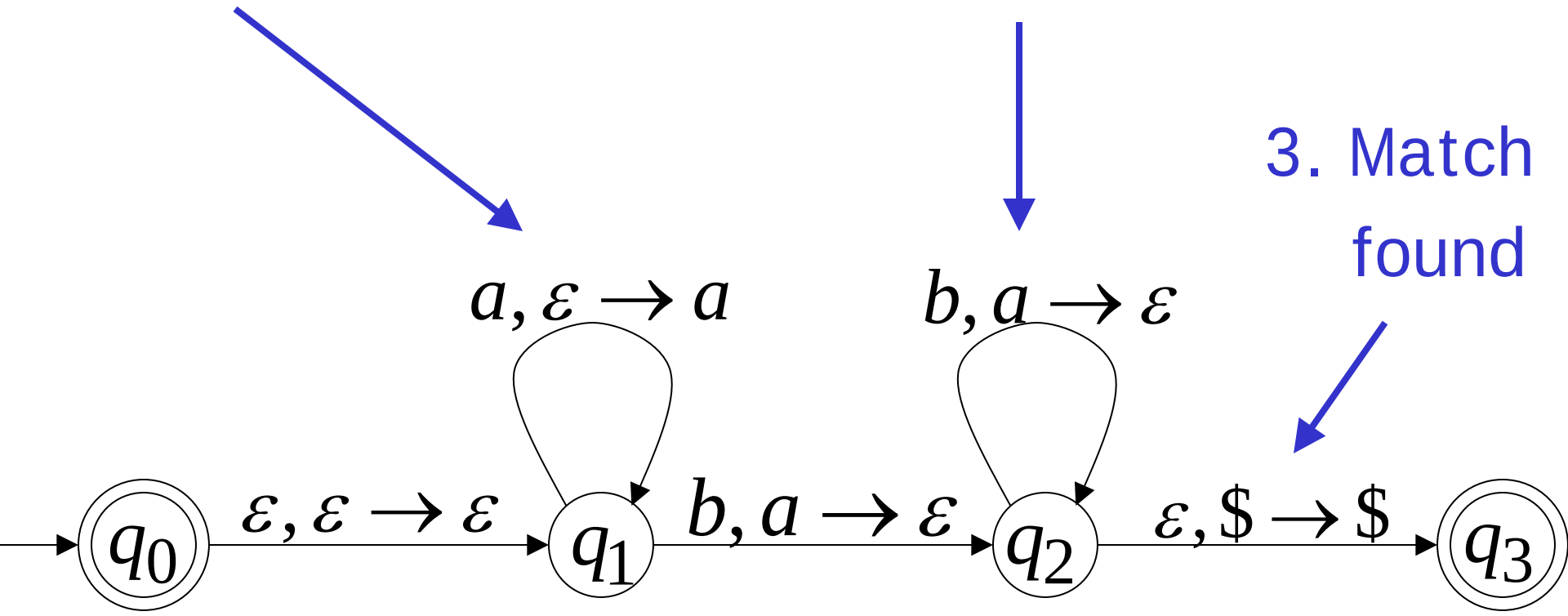
$$L(M) = \{a^n b^n : n \geq 0\}$$

Basic Idea:

1. Push the a's on the stack

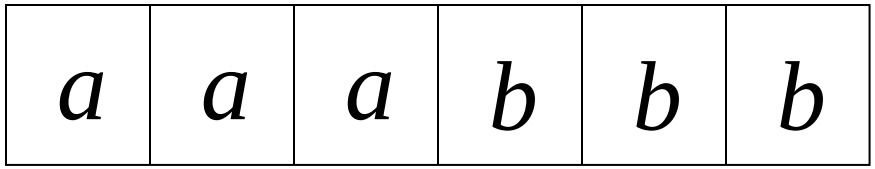
2. Match the b's on input with a's on stack

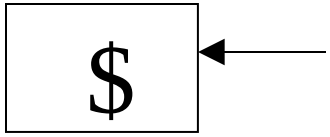
3. Match found



Execution Example: Time 0

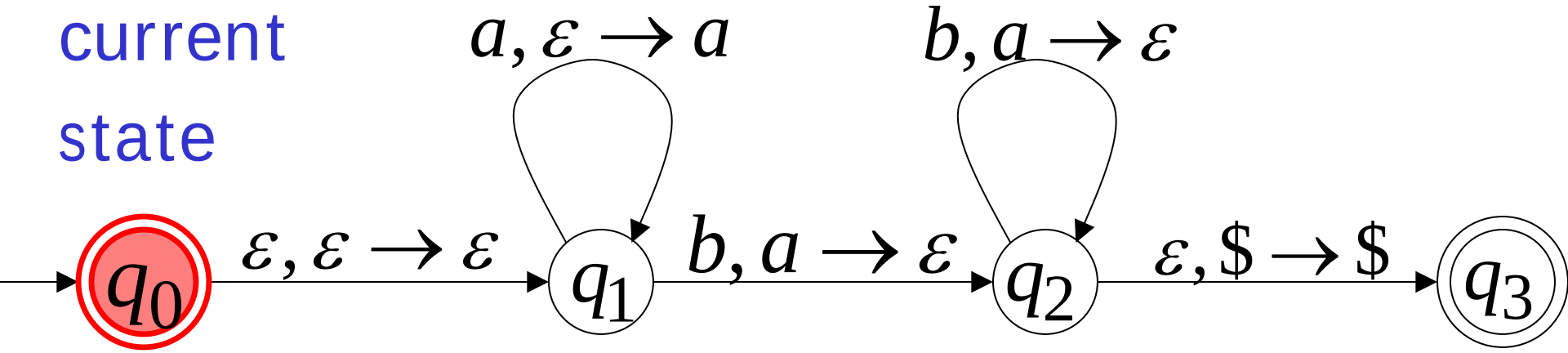
Input





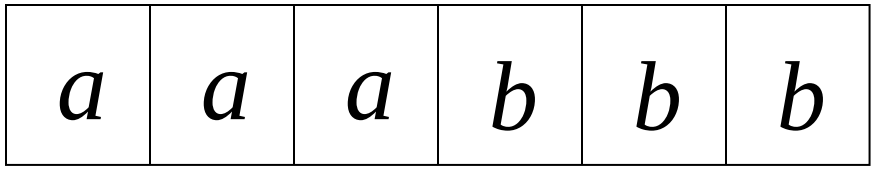
Stack

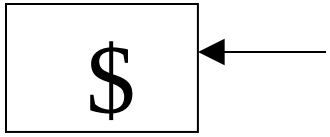
current state



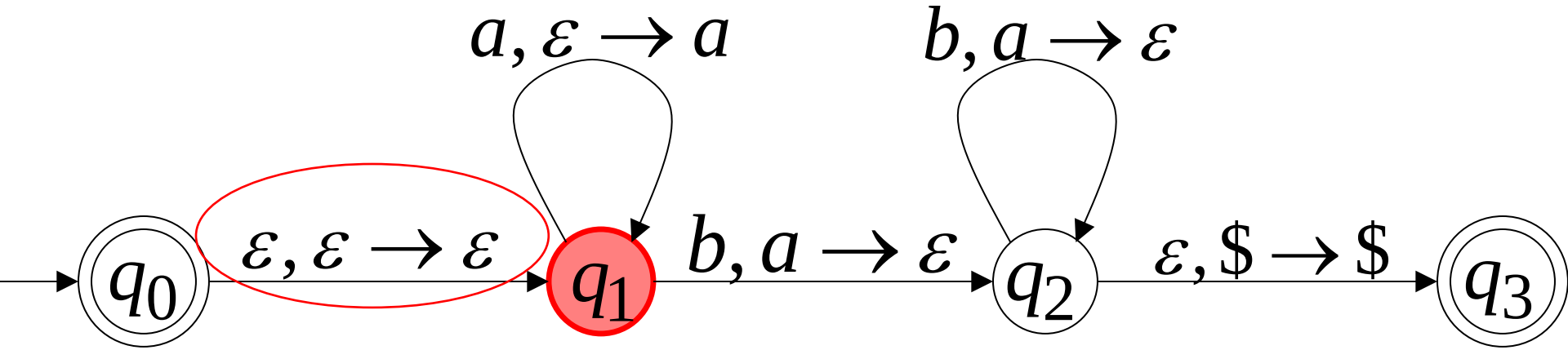
Time 1

Input



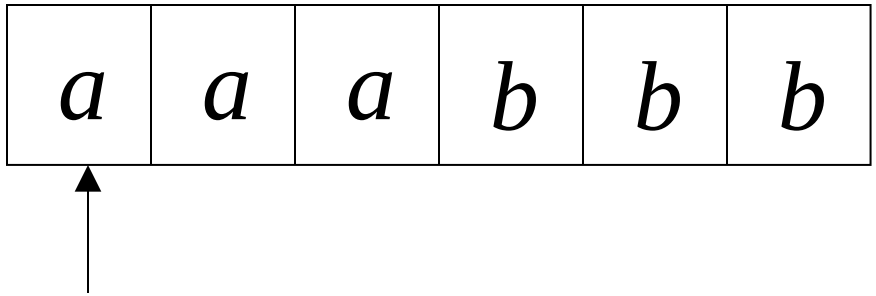


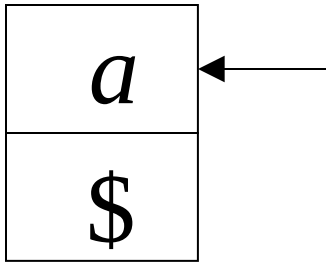
Stack



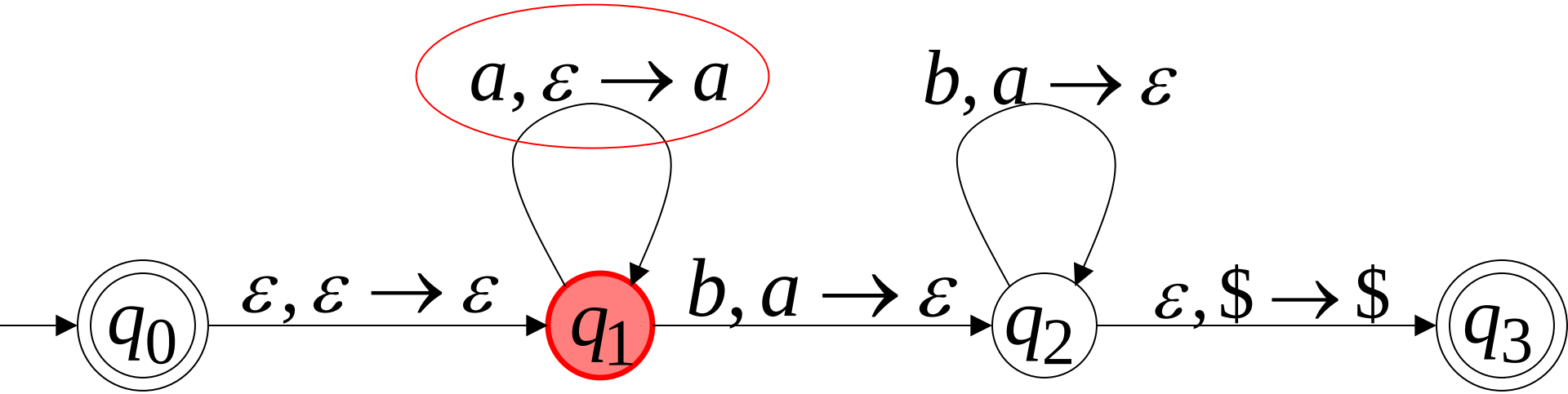
Time 2

Input



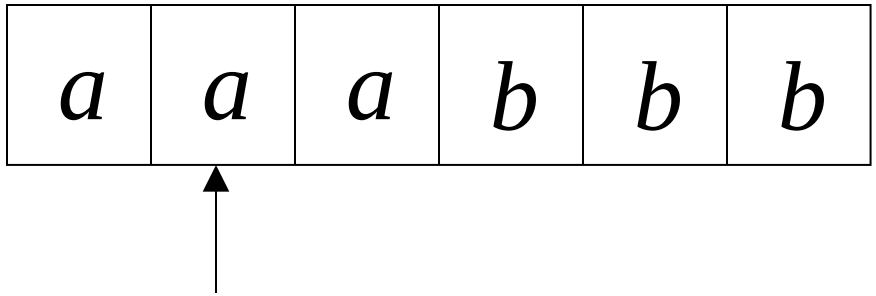


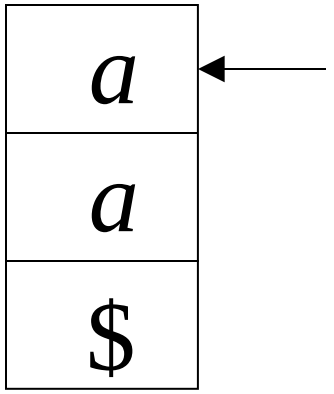
Stack



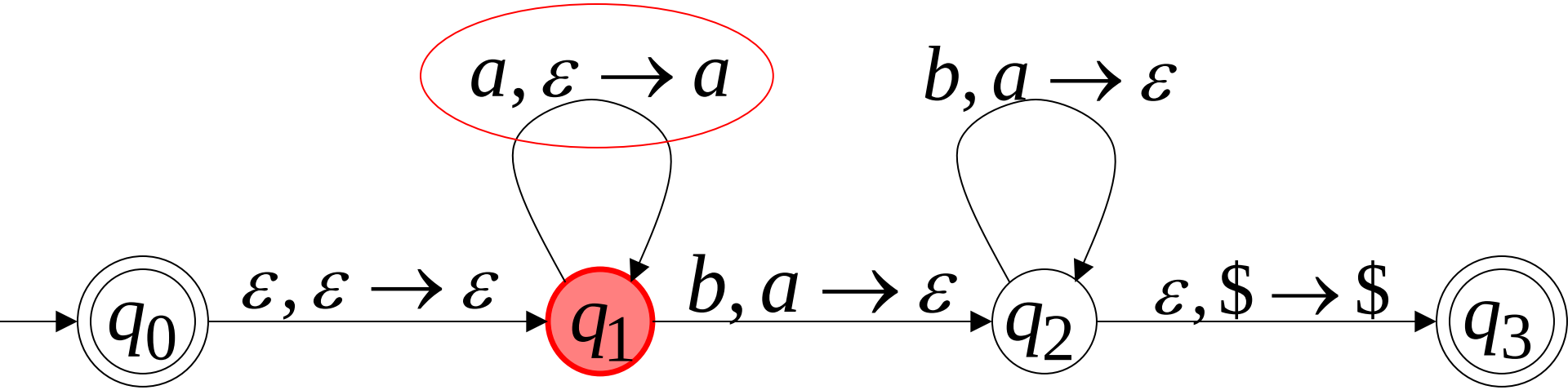
Time 3

Input



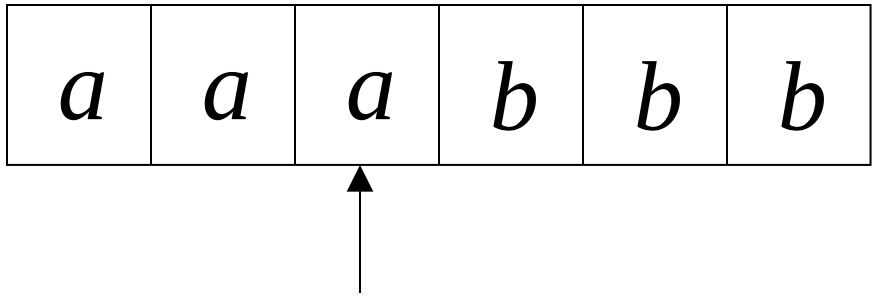


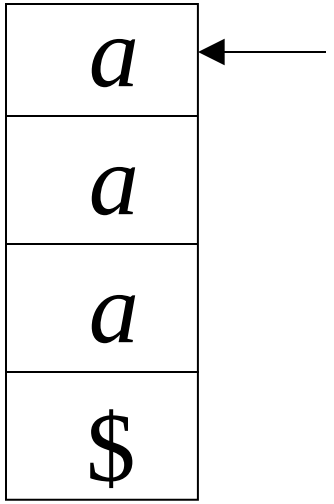
Stack



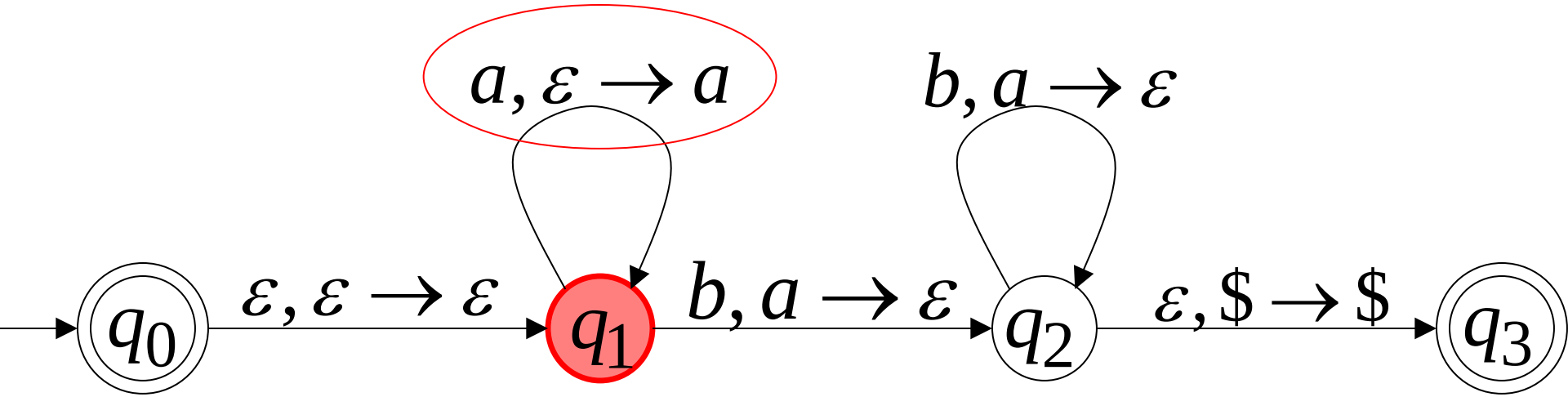
Time 4

Input



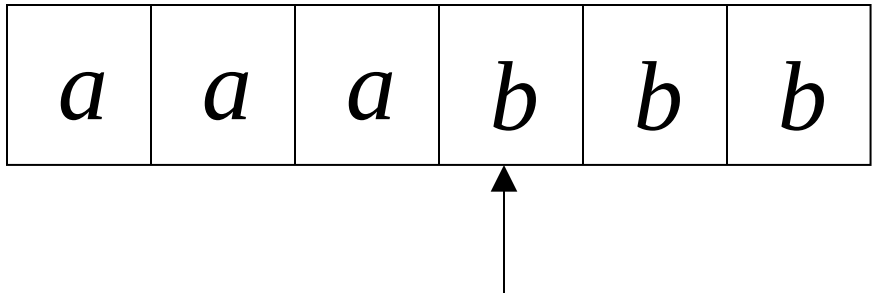


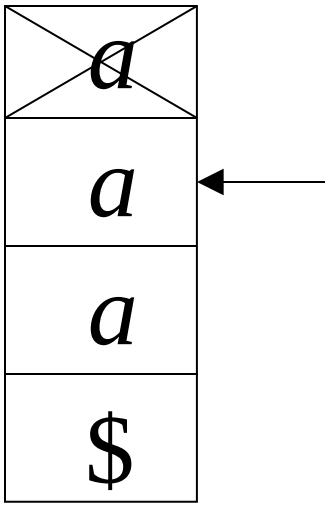
Stack



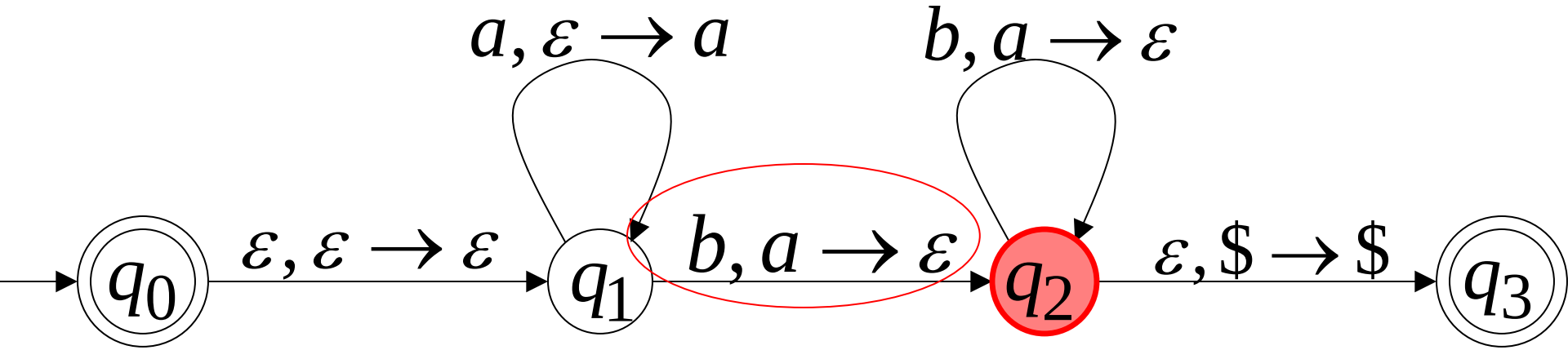
Time 5

Input



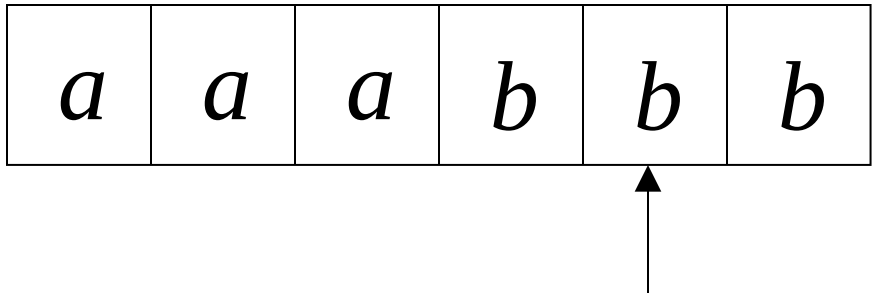


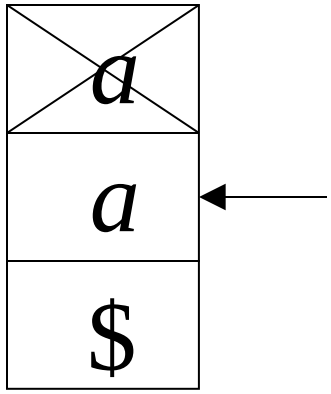
Stack



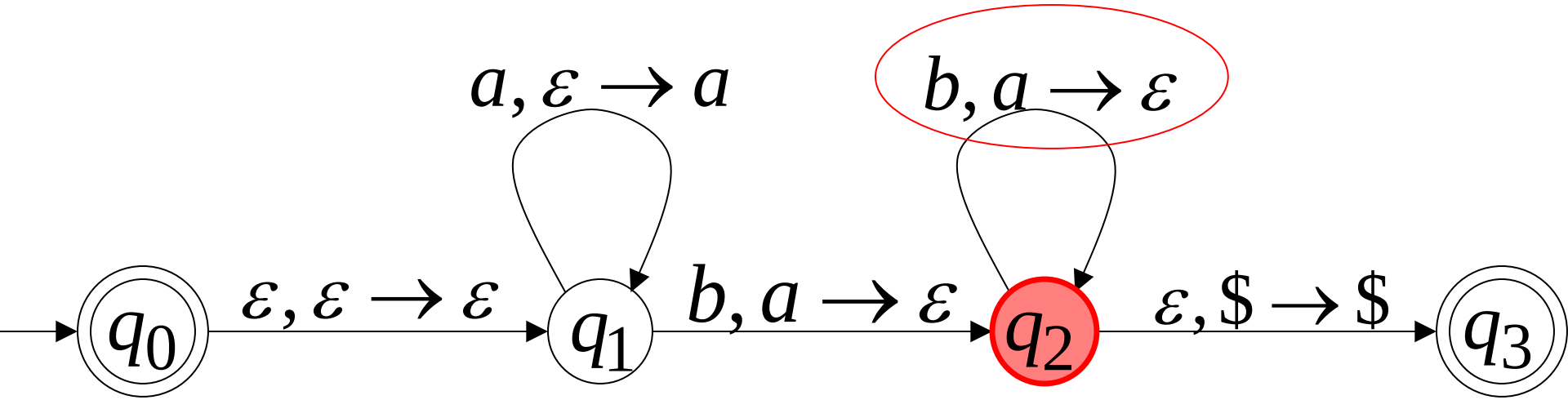
Time 6

Input



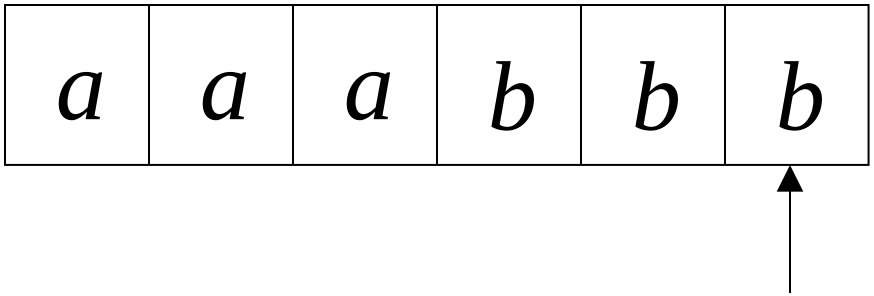


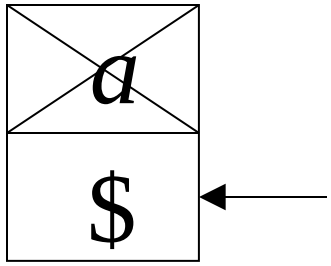
Stack



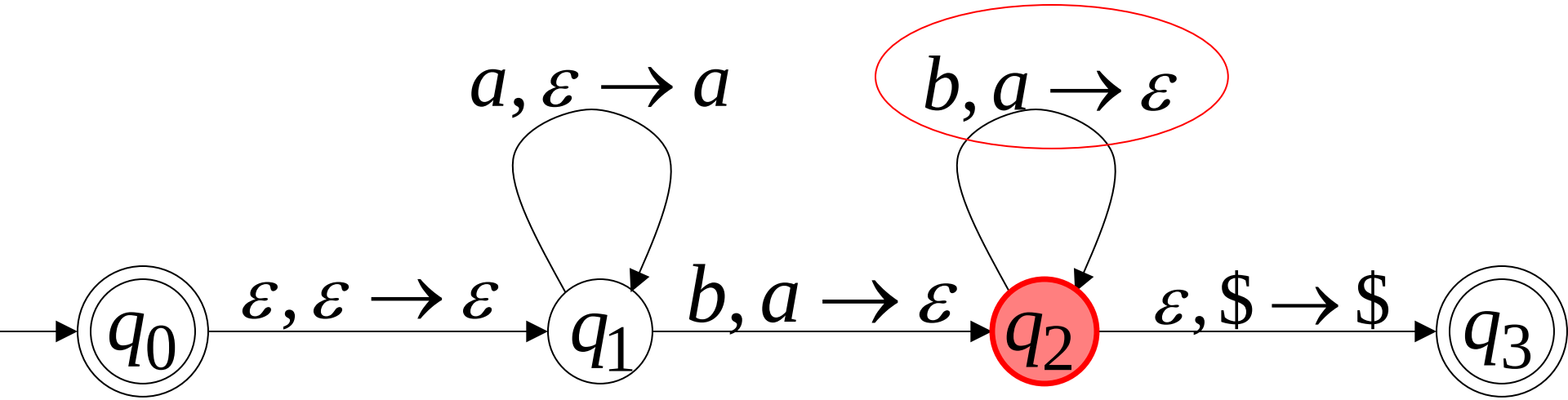
Time 7

Input



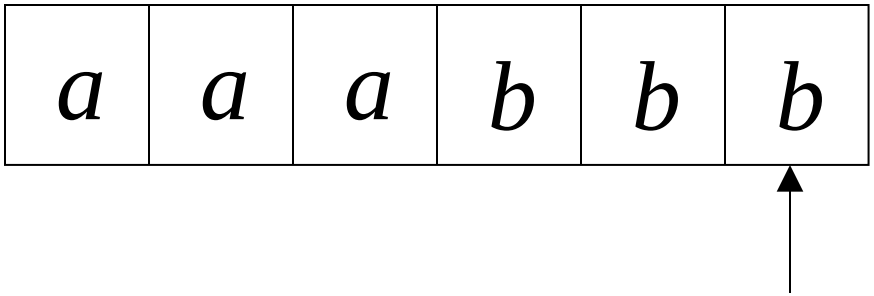


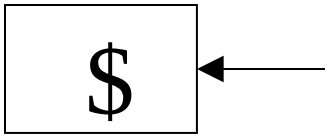
Stack



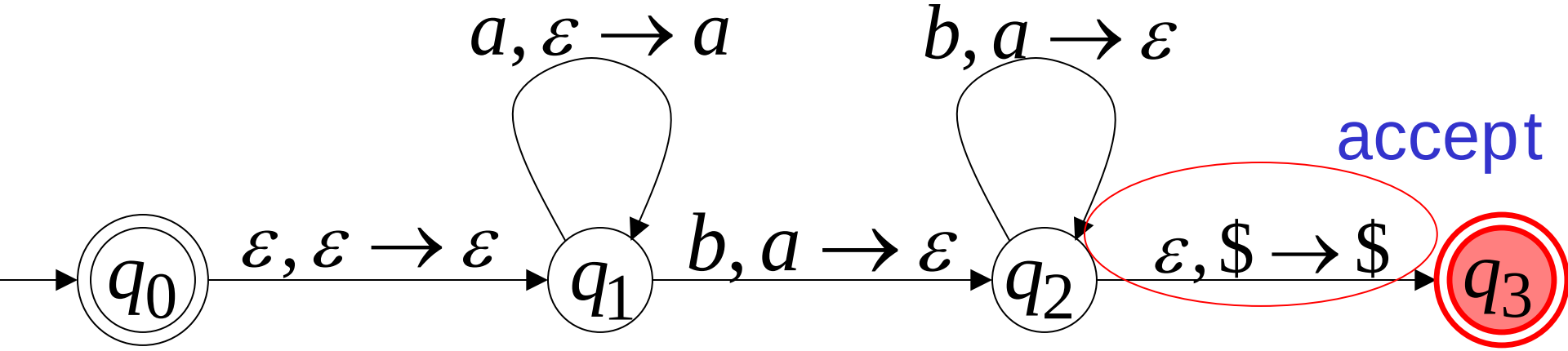
Time 8

Input





Stack



A string is accepted if there is
a computation such that:

All the input is consumed

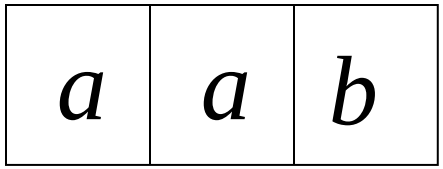
AND

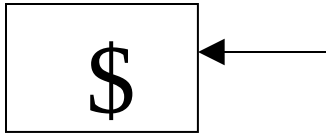
The last state is an accepting state

We do not care about the stack contents
at the end of the accepting computation

Rejection Example: Time 0

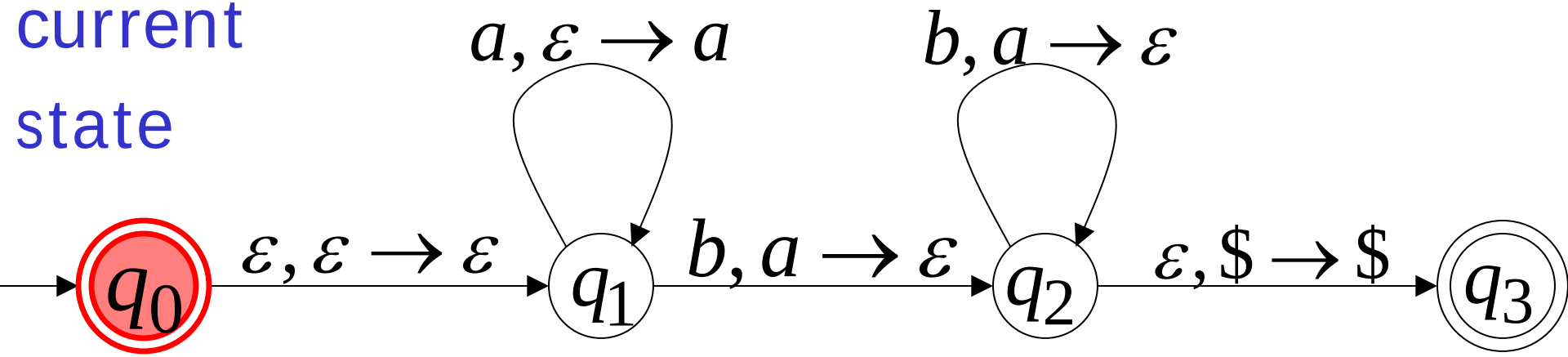
Input





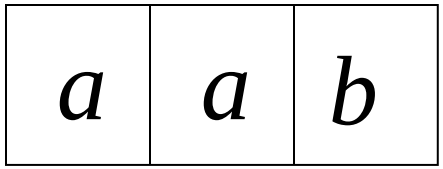
Stack

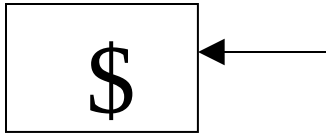
current state



Rejection Example: Time 1

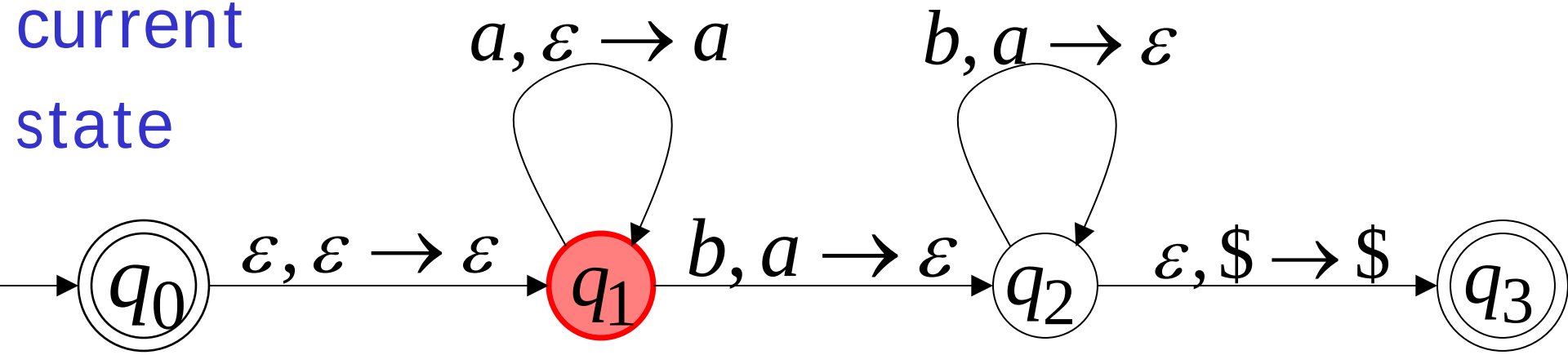
Input





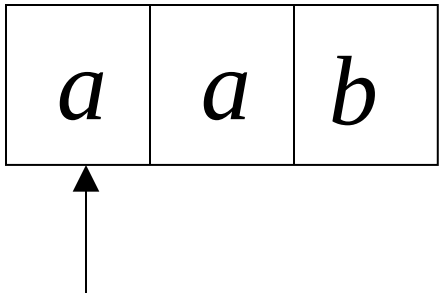
Stack

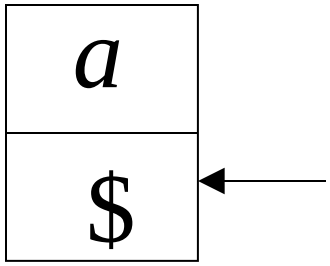
current state



Rejection Example: Time 2

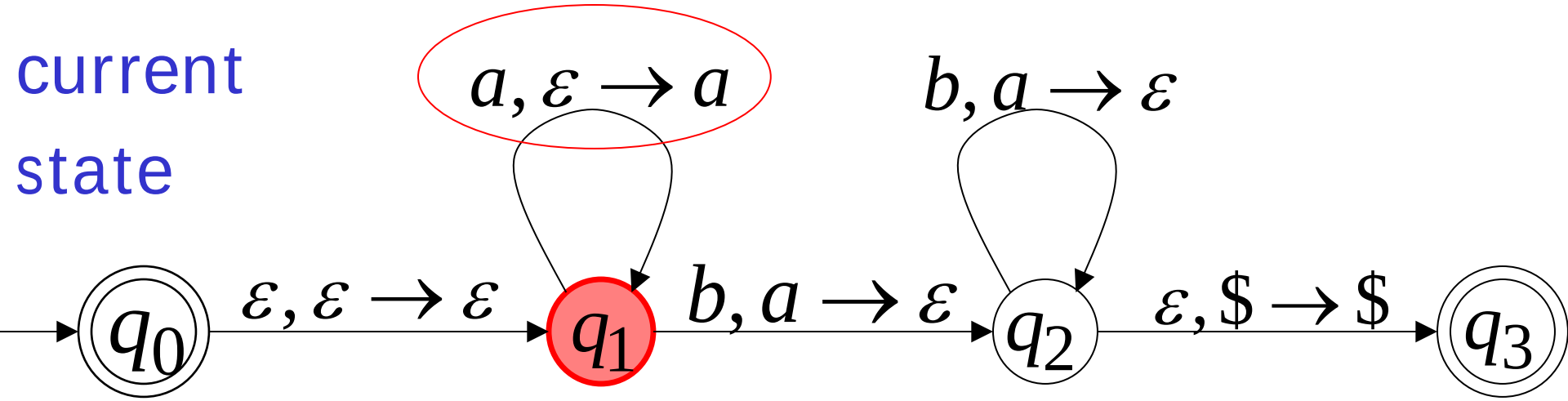
Input





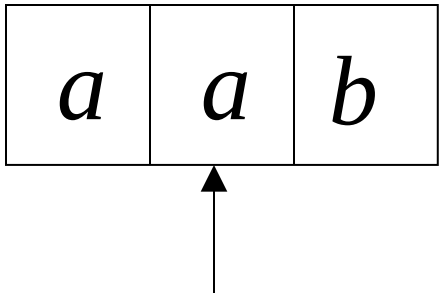
Stack

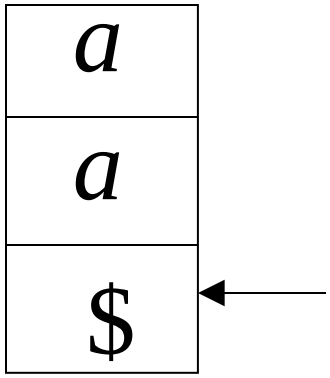
current state



Rejection Example: Time 3

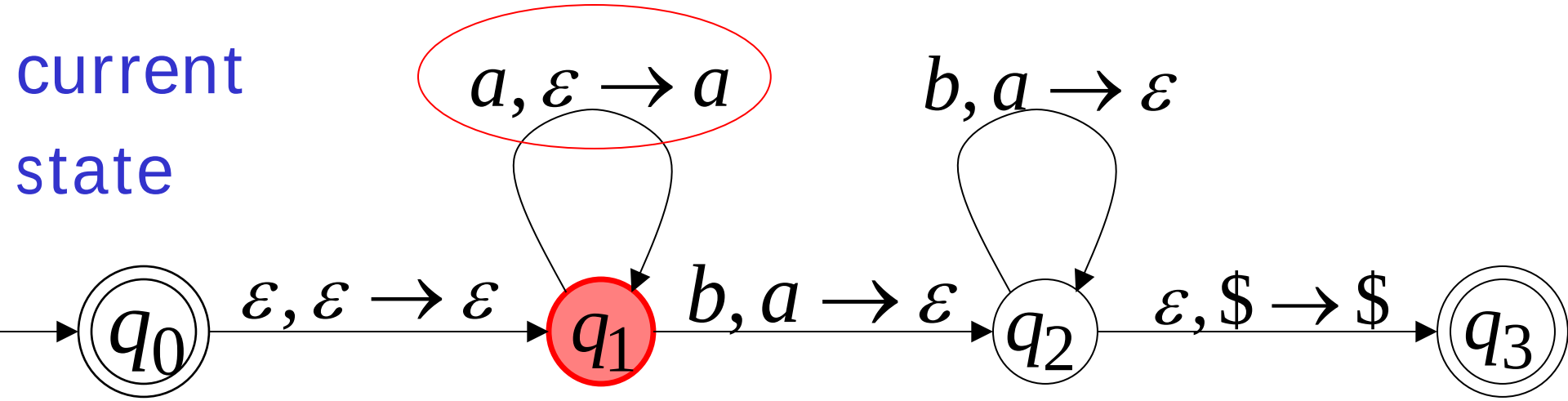
Input





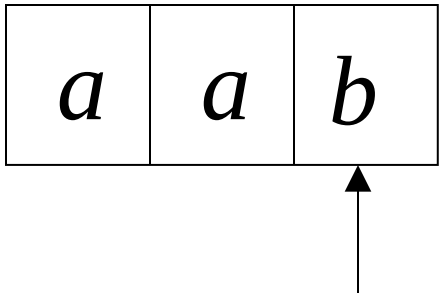
Stack

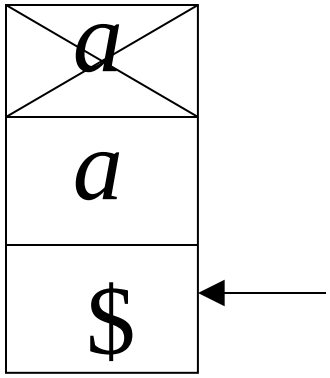
current state



Rejection Example: Time 4

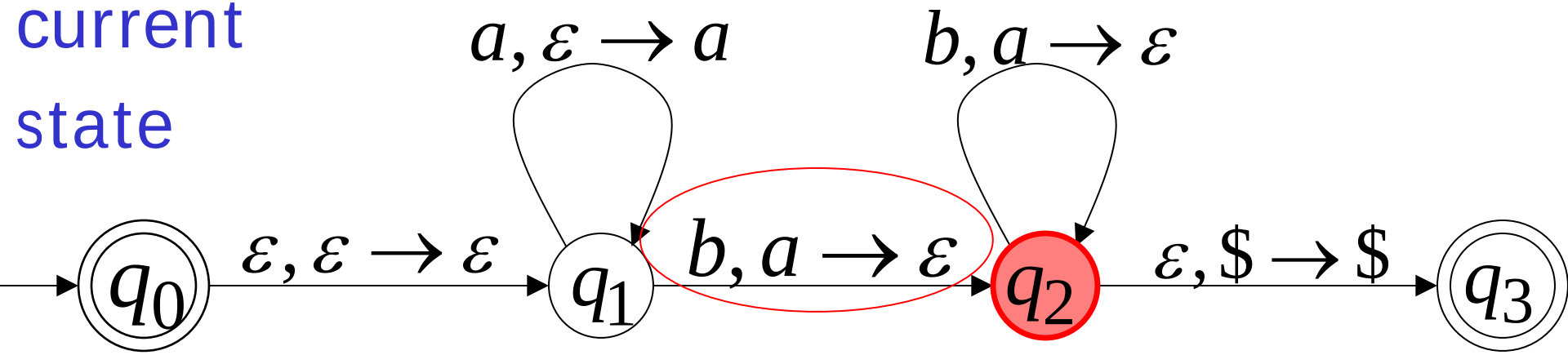
Input





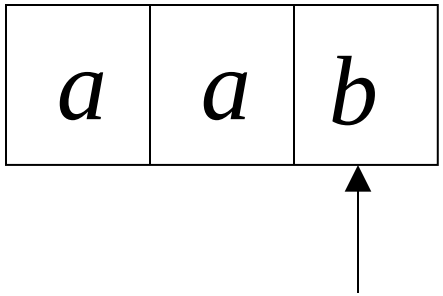
Stack

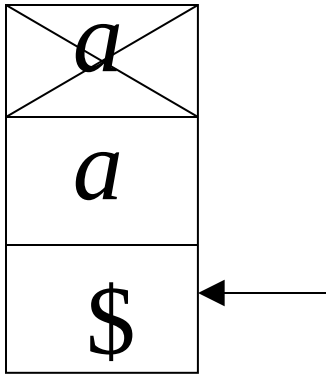
current state



Rejection Example: Time 4

Input

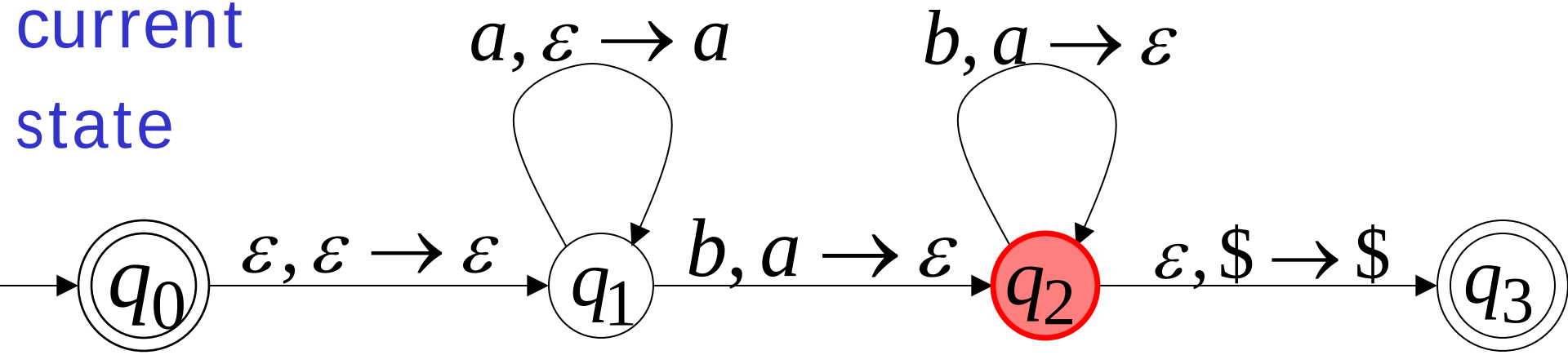




Stack

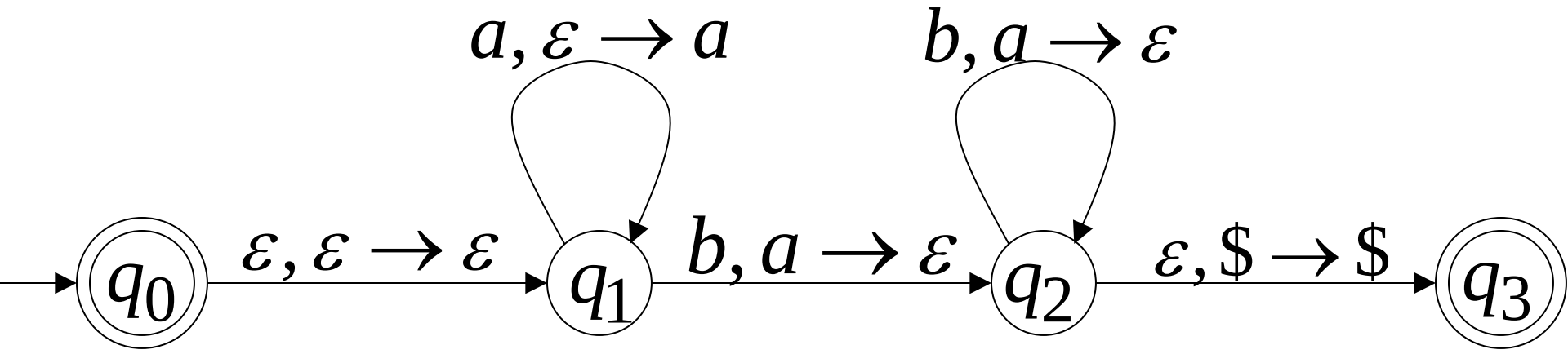
reject

current state



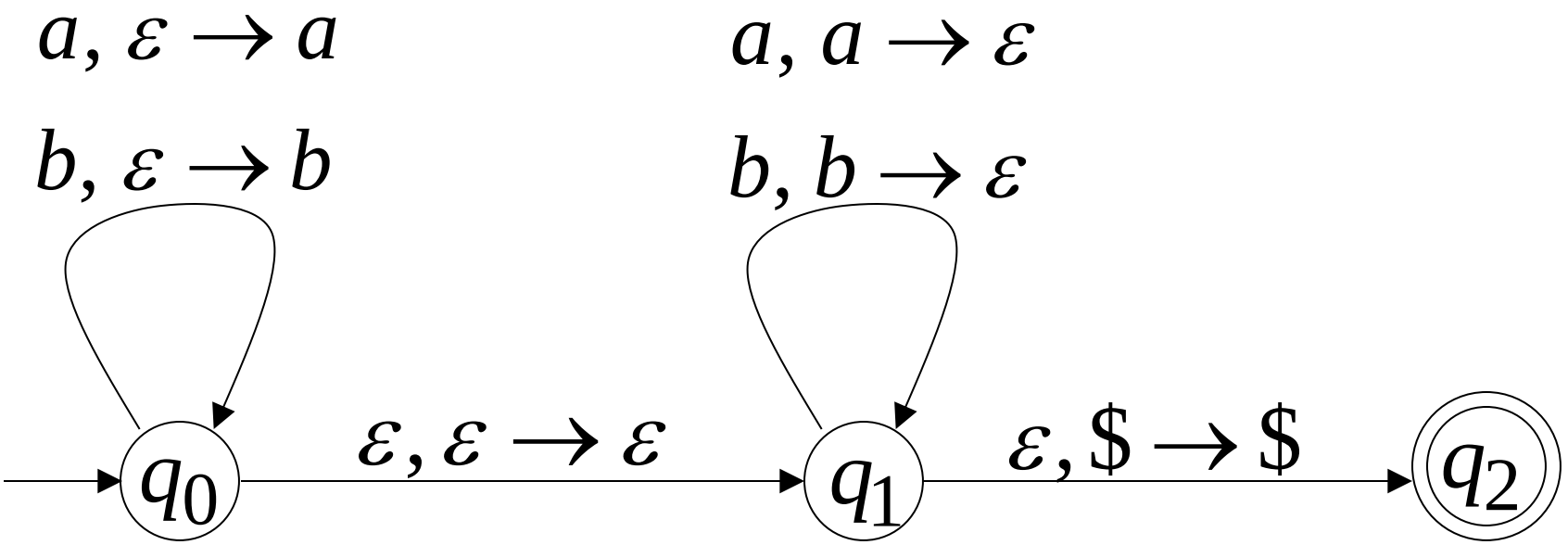
There is no accepting computation for aab

The string aab is rejected by the PDA



Another PDA example

PDA M : $L(M) = \{vv^R : v \in \{a,b\}^*\}$



Basic Idea:

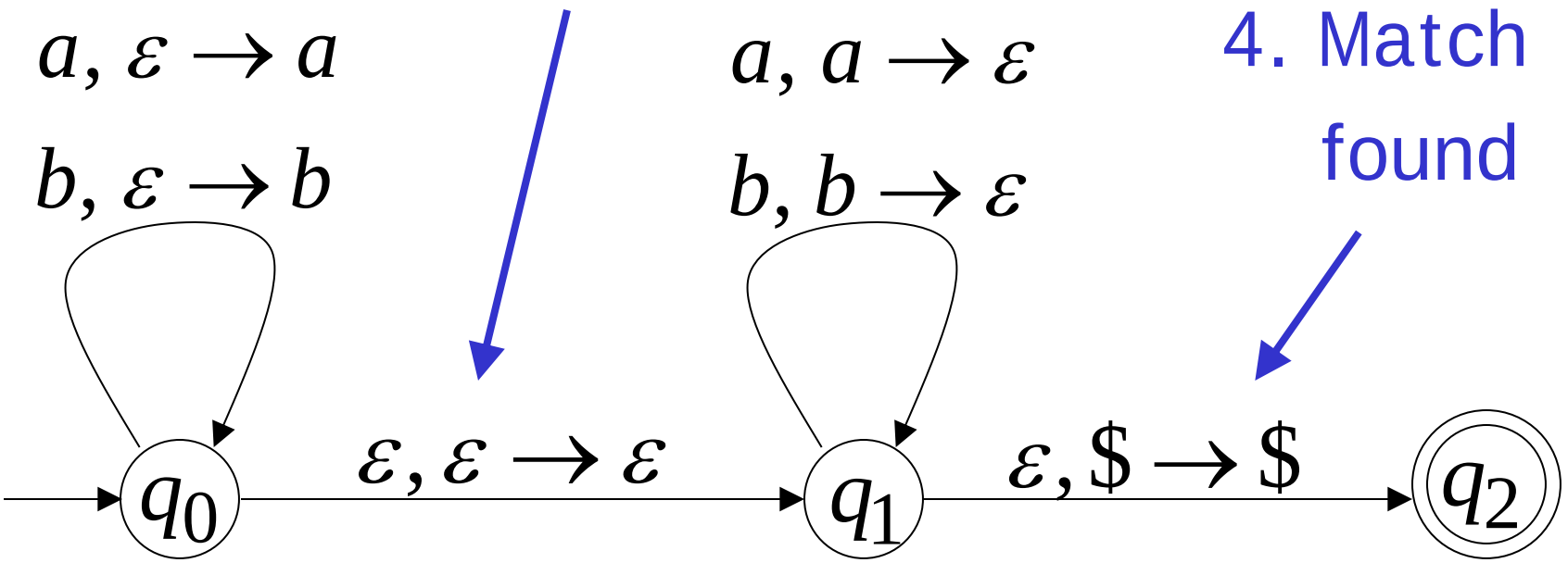
$$L(M) = \{vv^R : v \in \{a,b\}^*\}$$

1. Push v on stack

2. Guess middle of input

3. Match v^R on input with v on stack

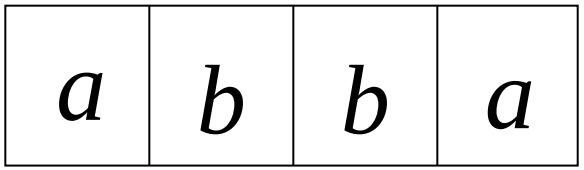
4. Match found

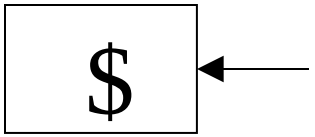


Execution Example:

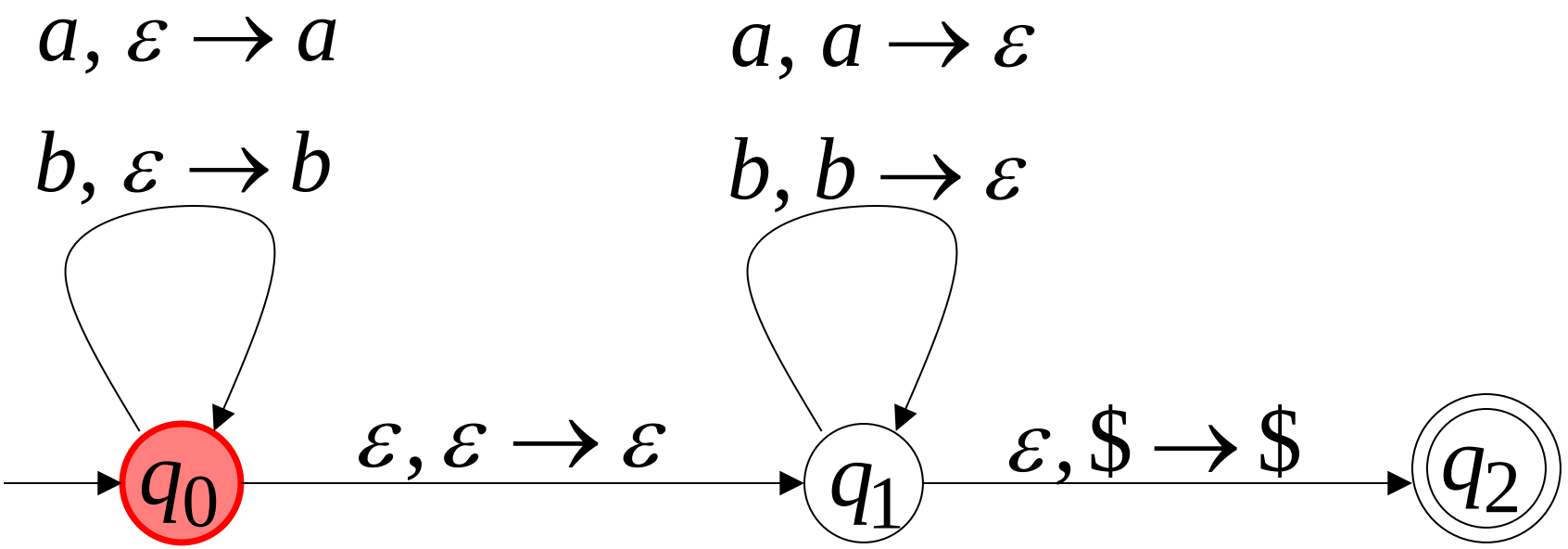
Time 0

Input



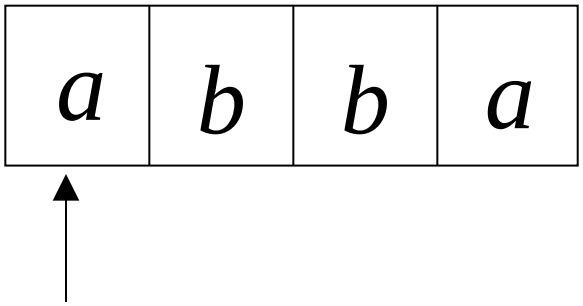


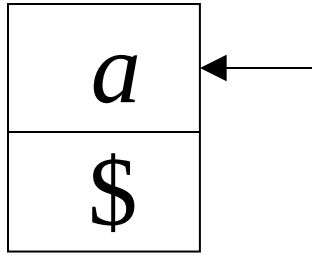
Stack



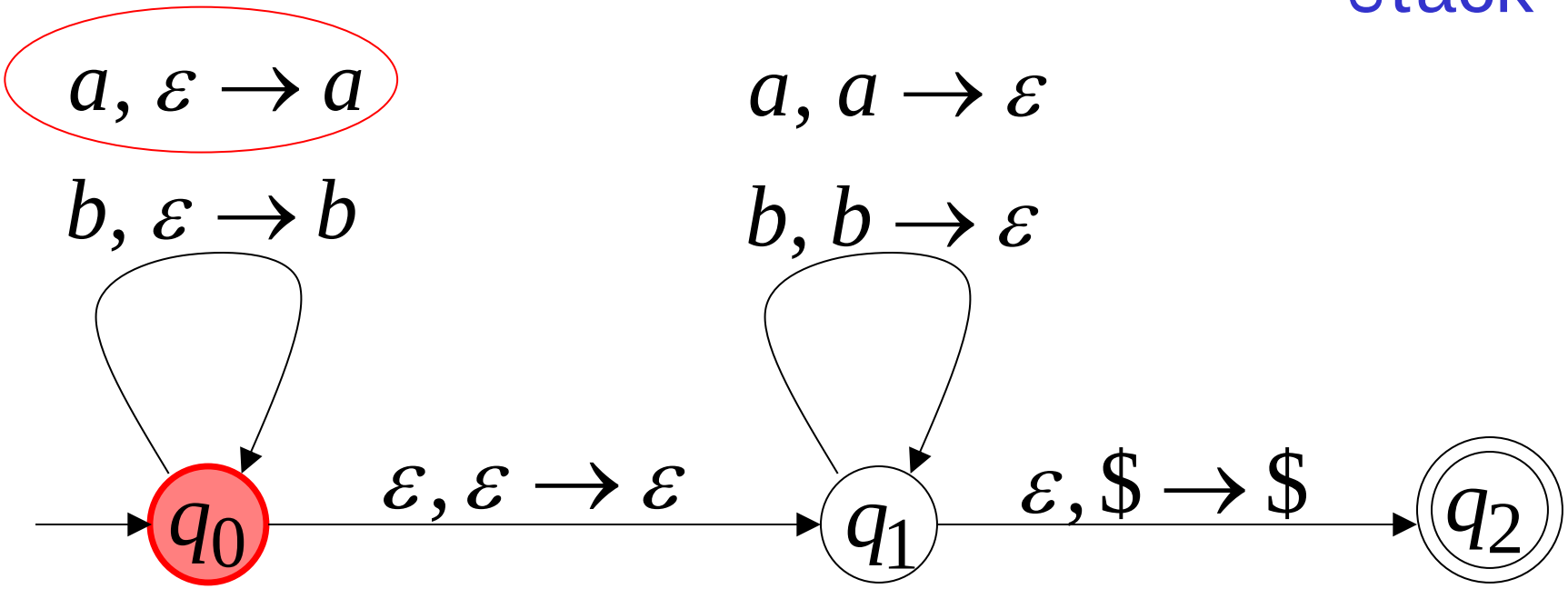
Time 1

Input



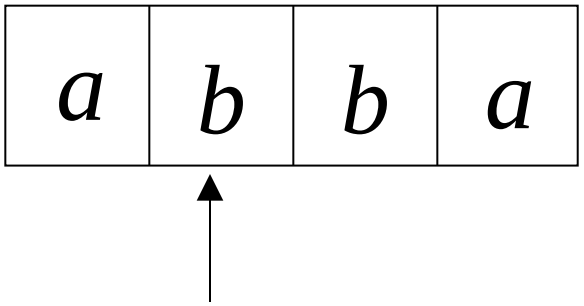


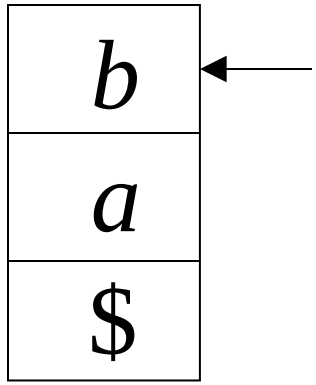
Stack



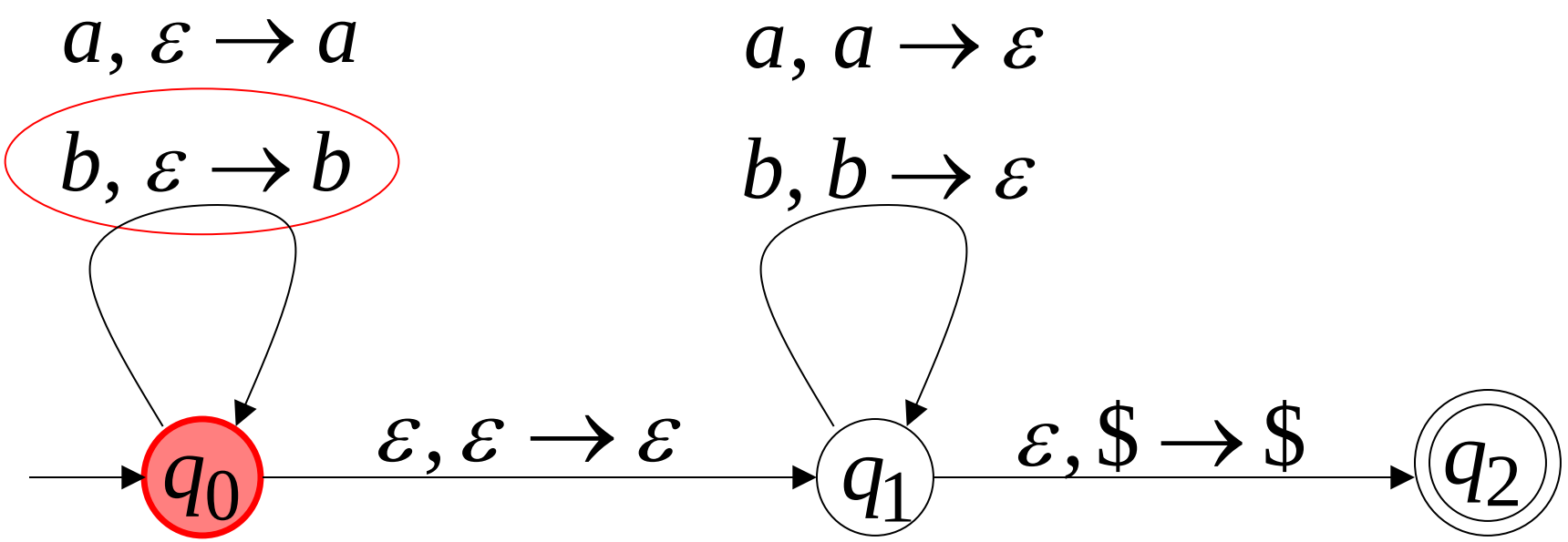
Time 2

Input



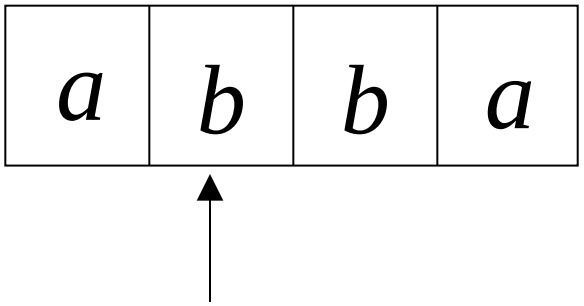


Stack

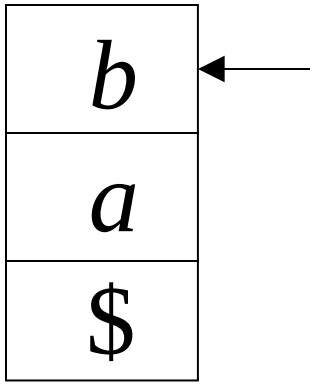


Time 3

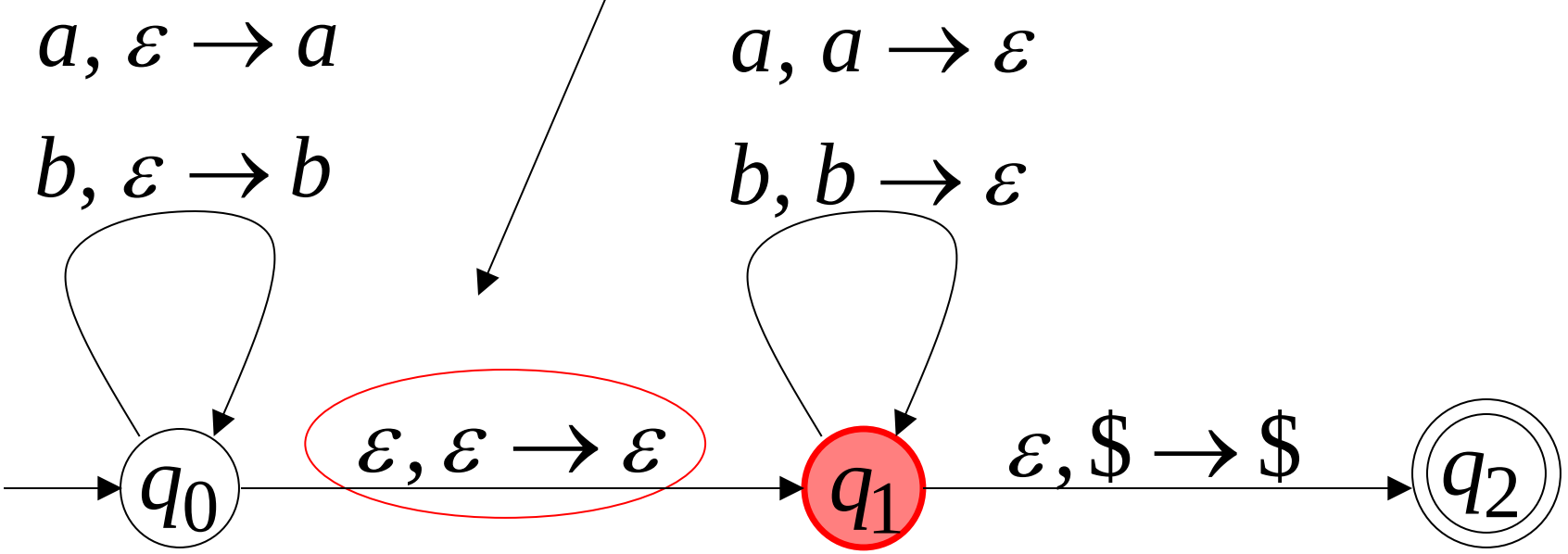
Input



Guess the middle of string

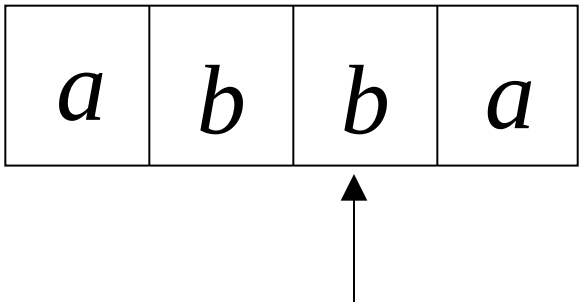


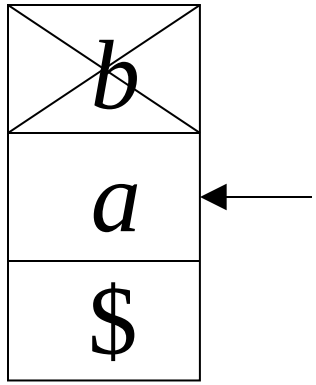
Stack



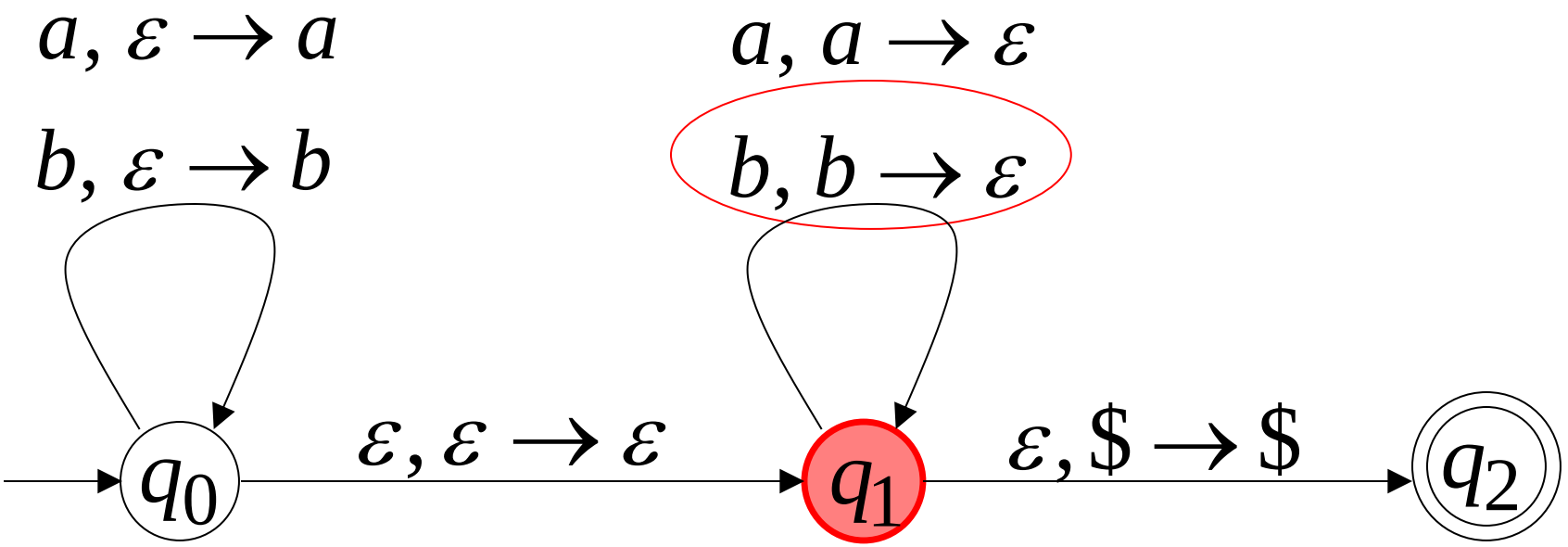
Time 4

Input



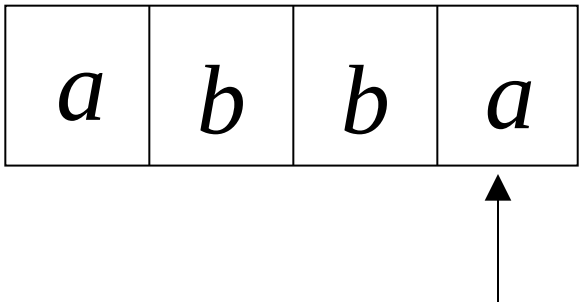


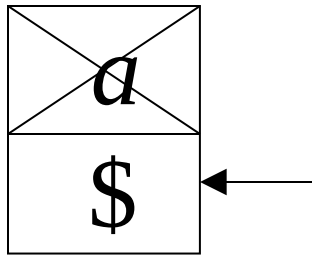
Stack



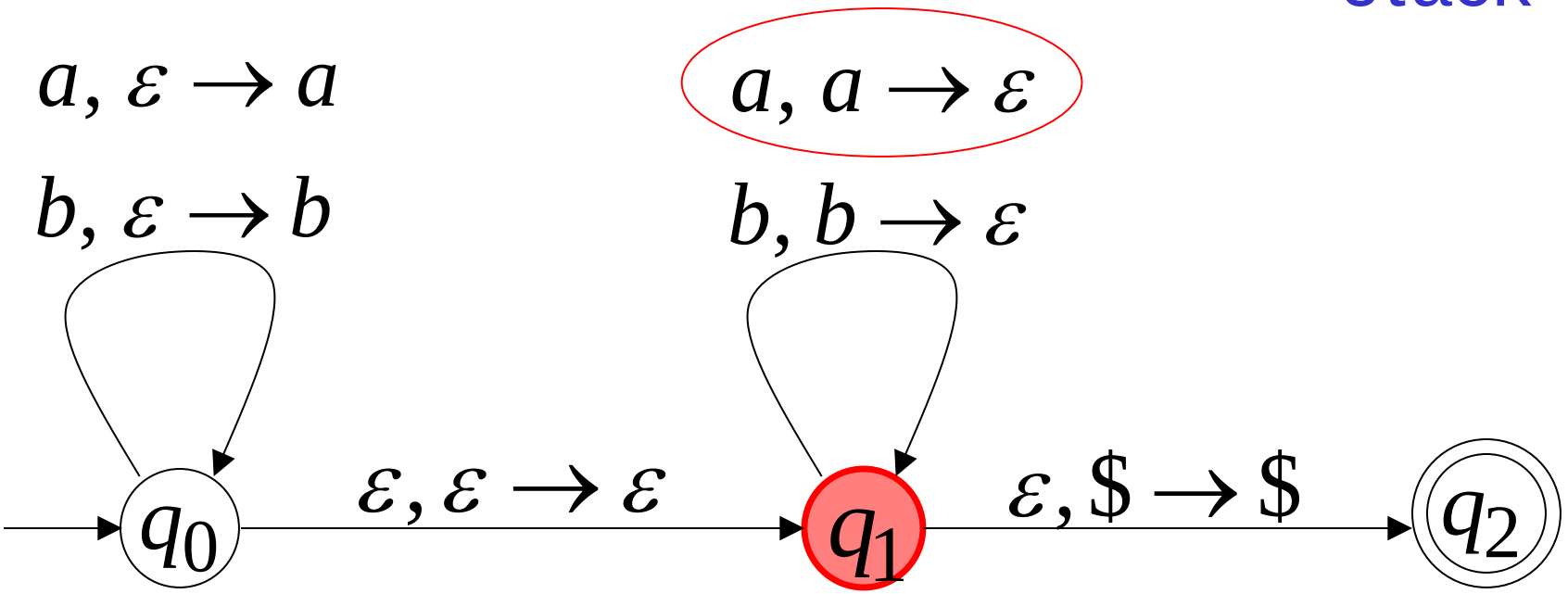
Time 5

Input



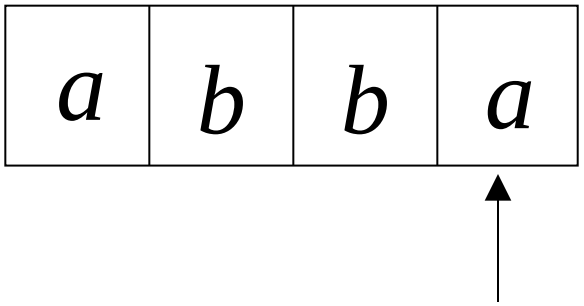


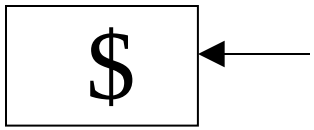
Stack



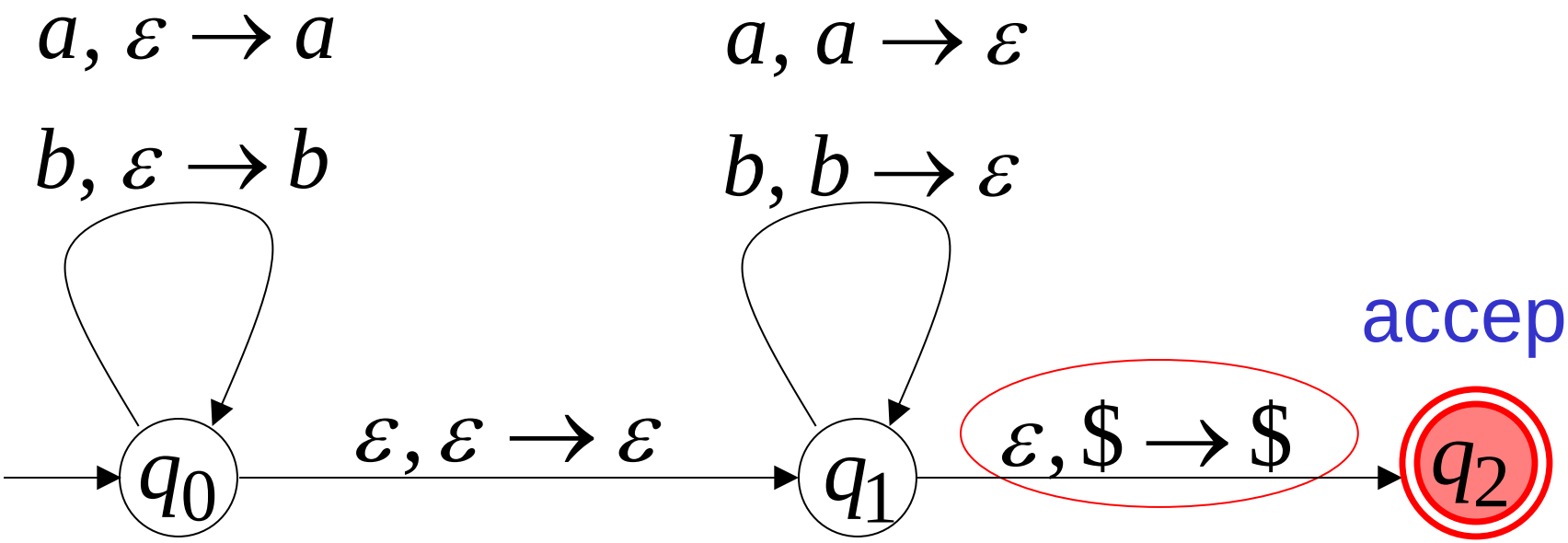
Time 6

Input





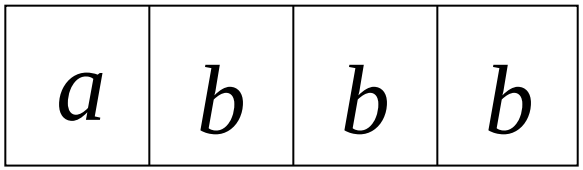
Stack

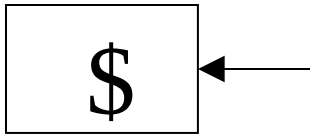


Rejection Example:

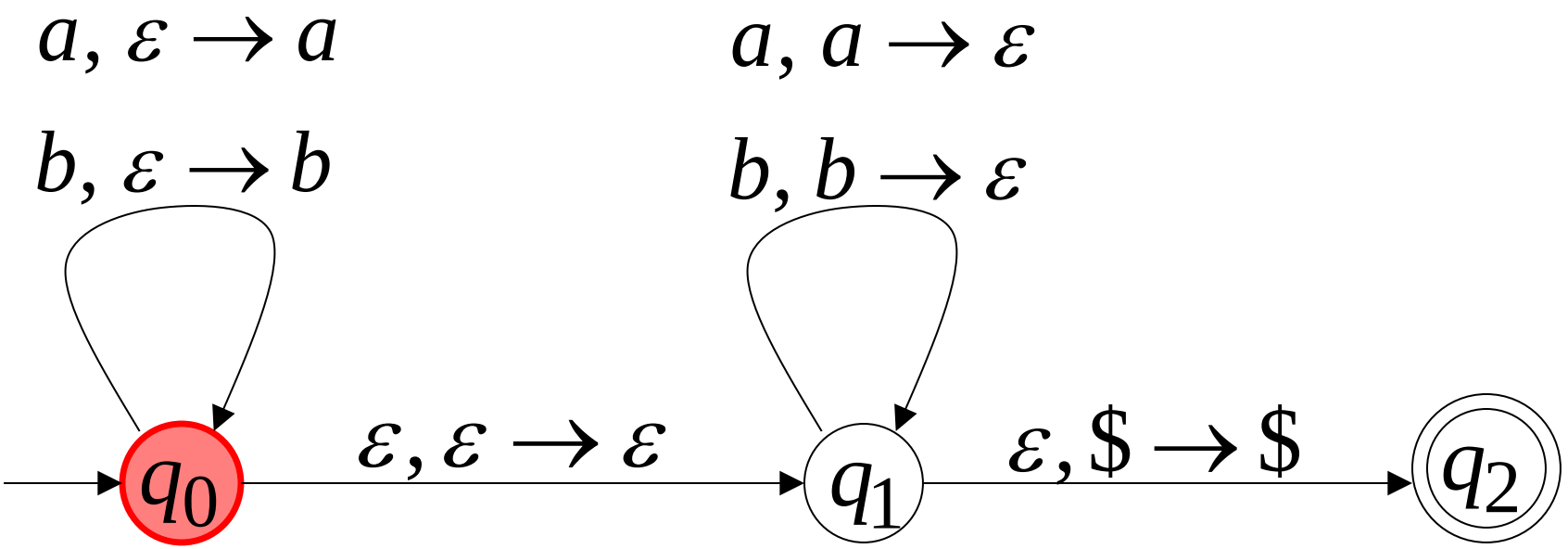
Time 0

Input



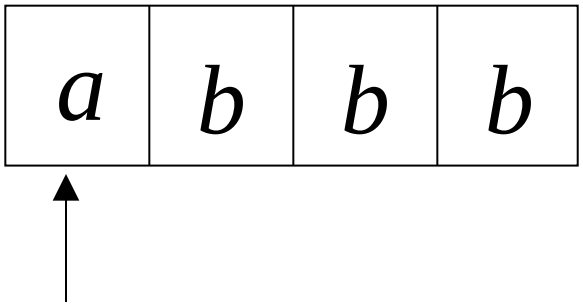


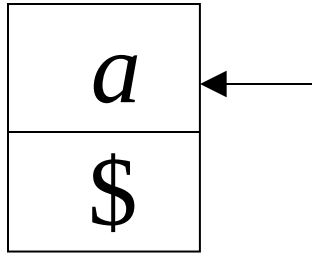
Stack



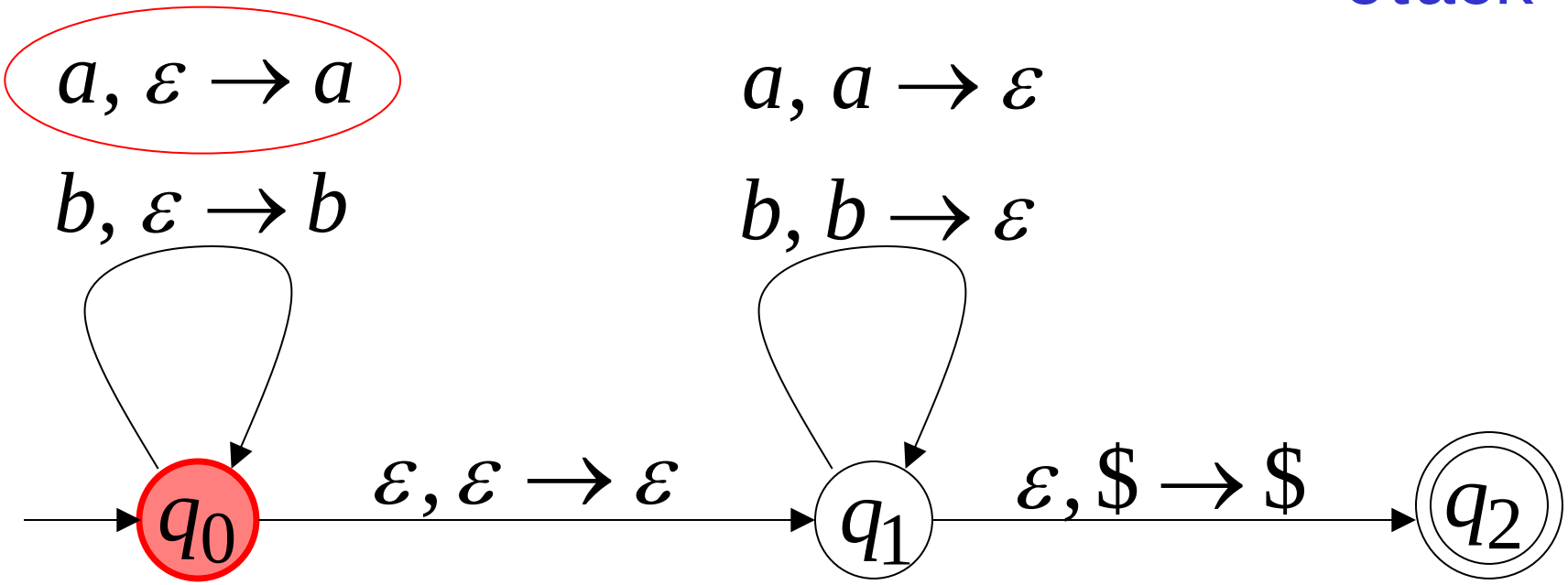
Time 1

Input



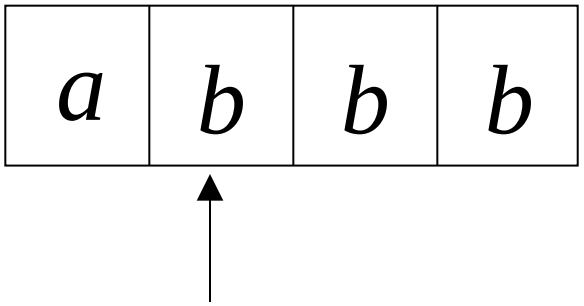


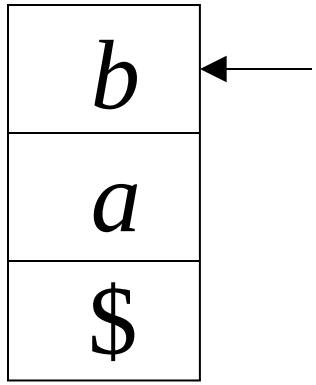
Stack



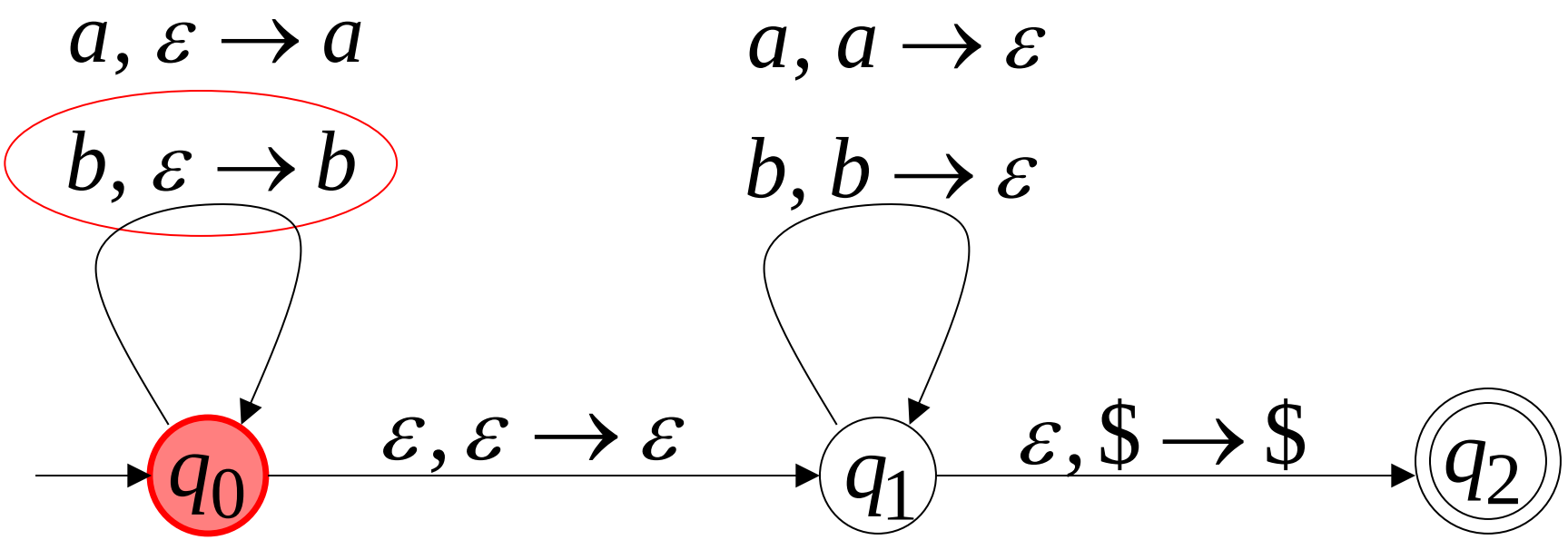
Time 2

Input



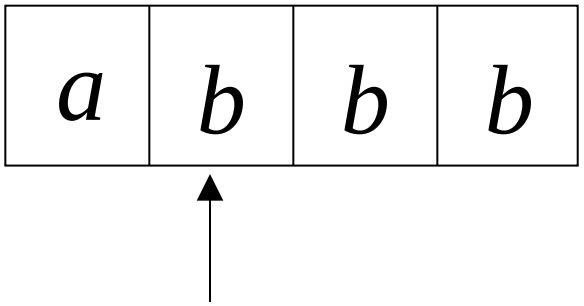


Stack

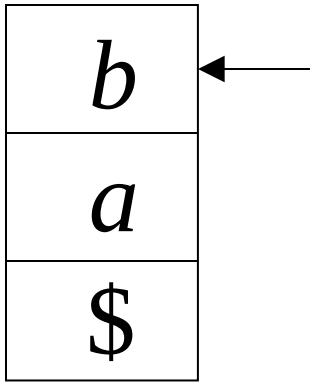


Time 3

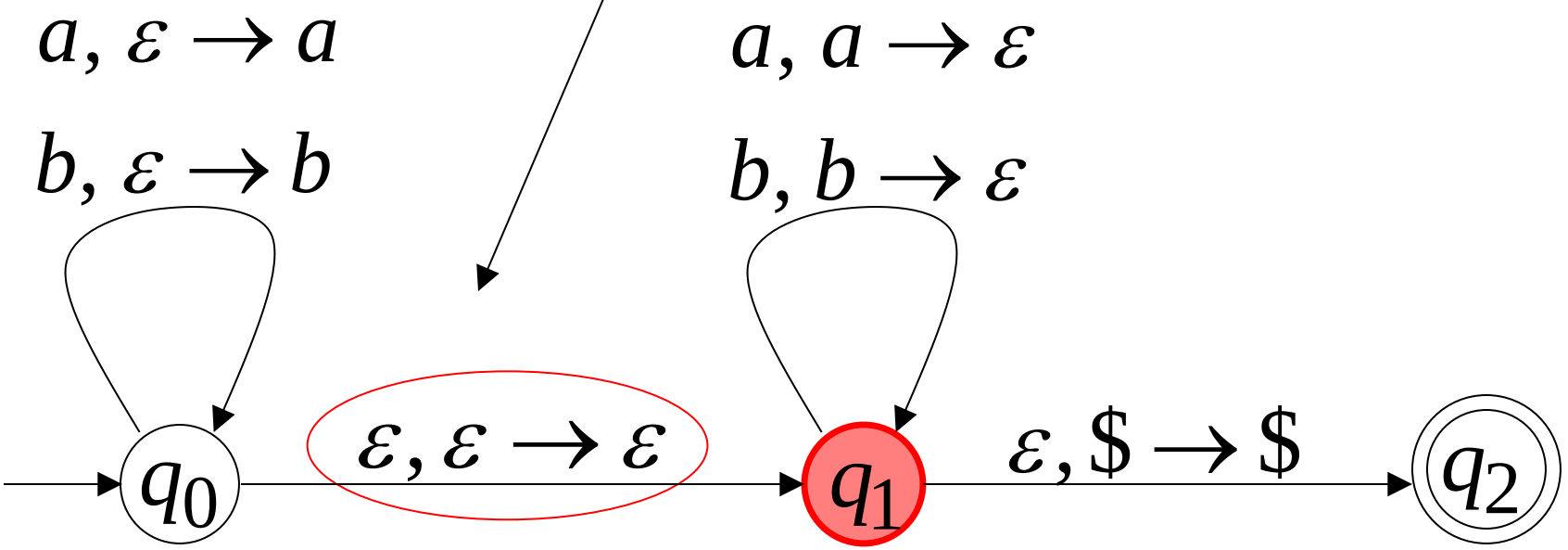
Input



Guess the middle of string

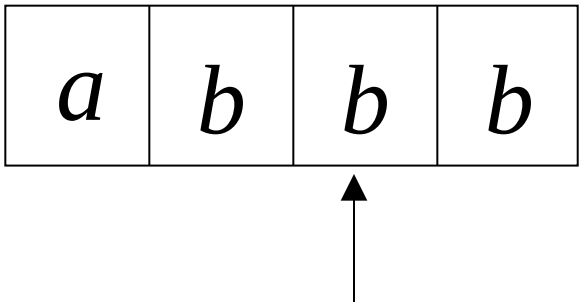


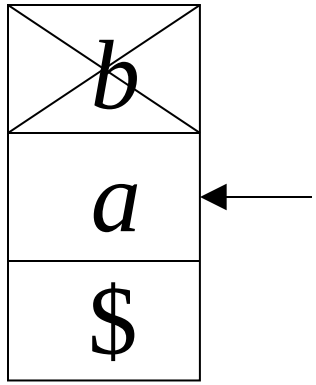
Stack



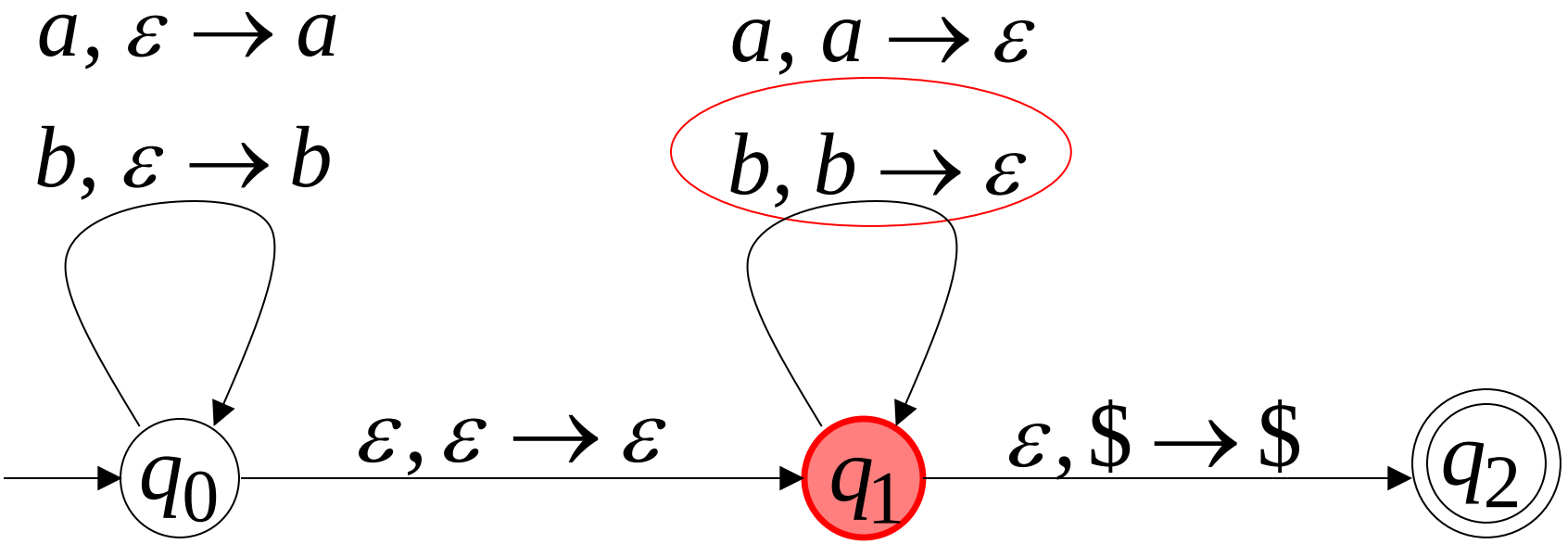
Time 4

Input





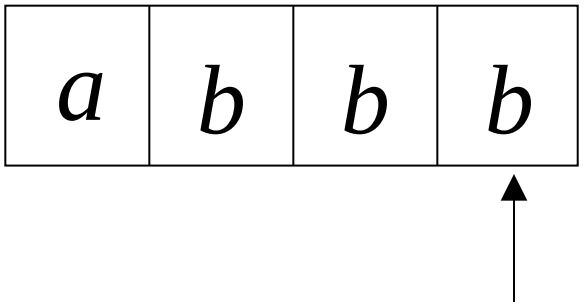
Stack



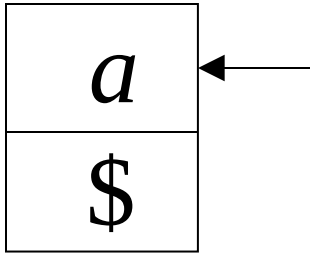
Time 5

There is no possible transition.

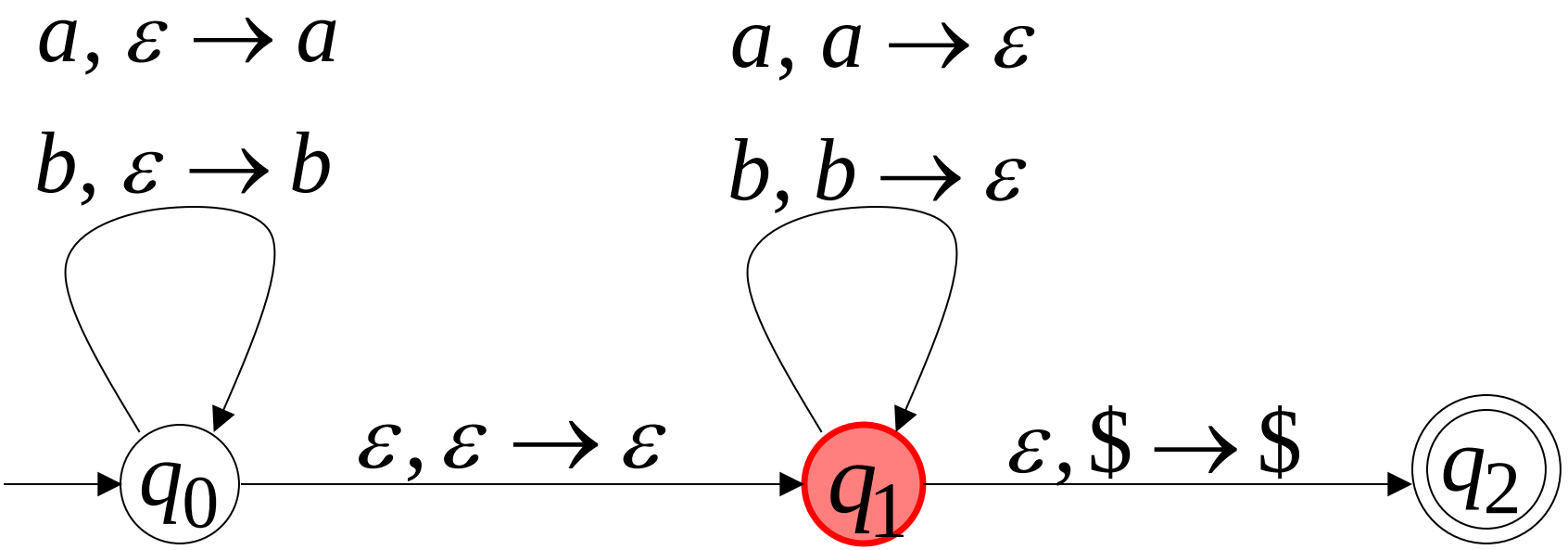
Input



Input is not consumed



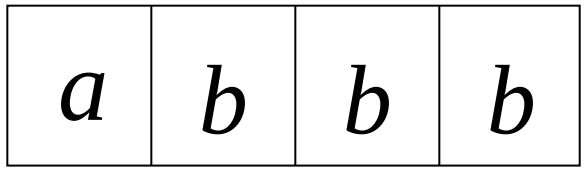
Stack



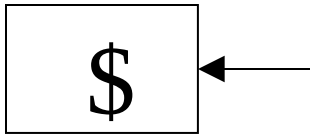
Another computation on same string:

Input

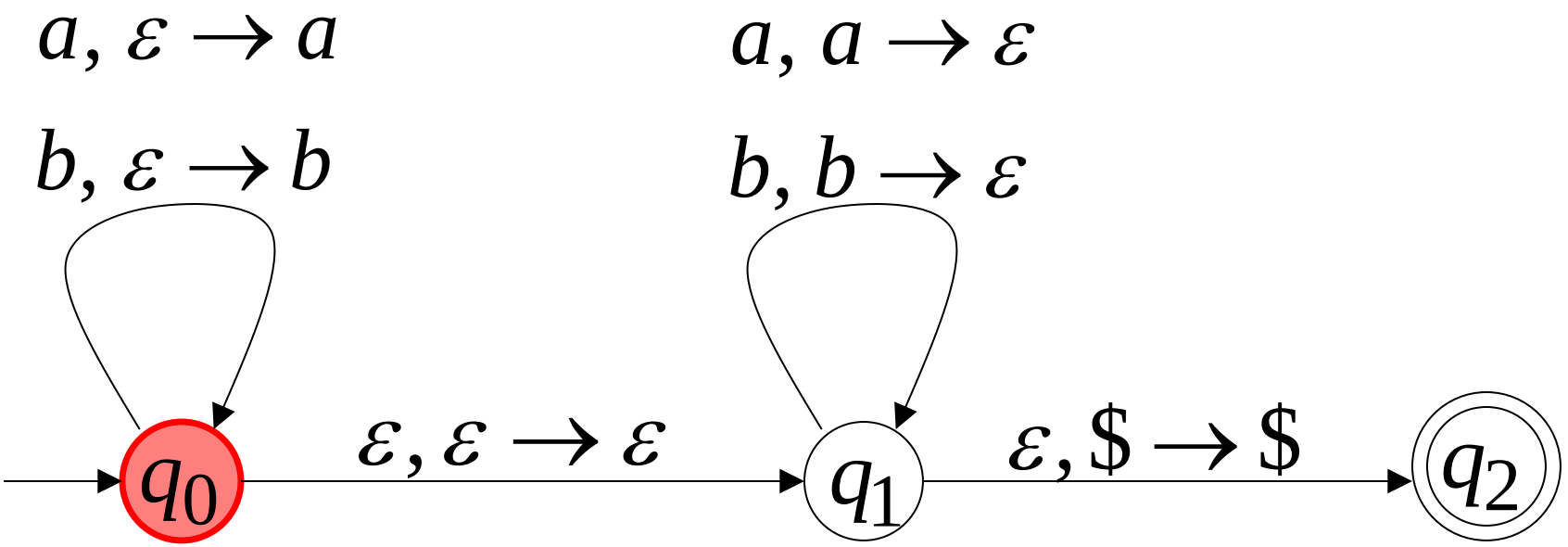
Time 0





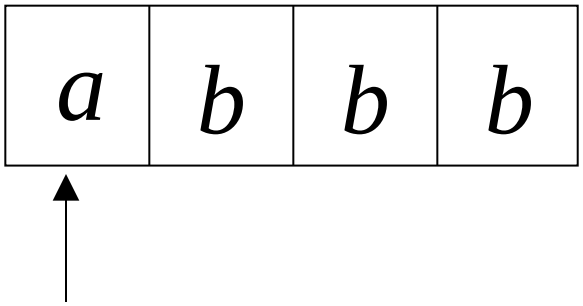


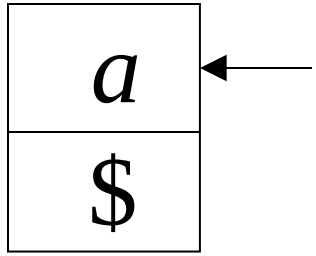
Stack



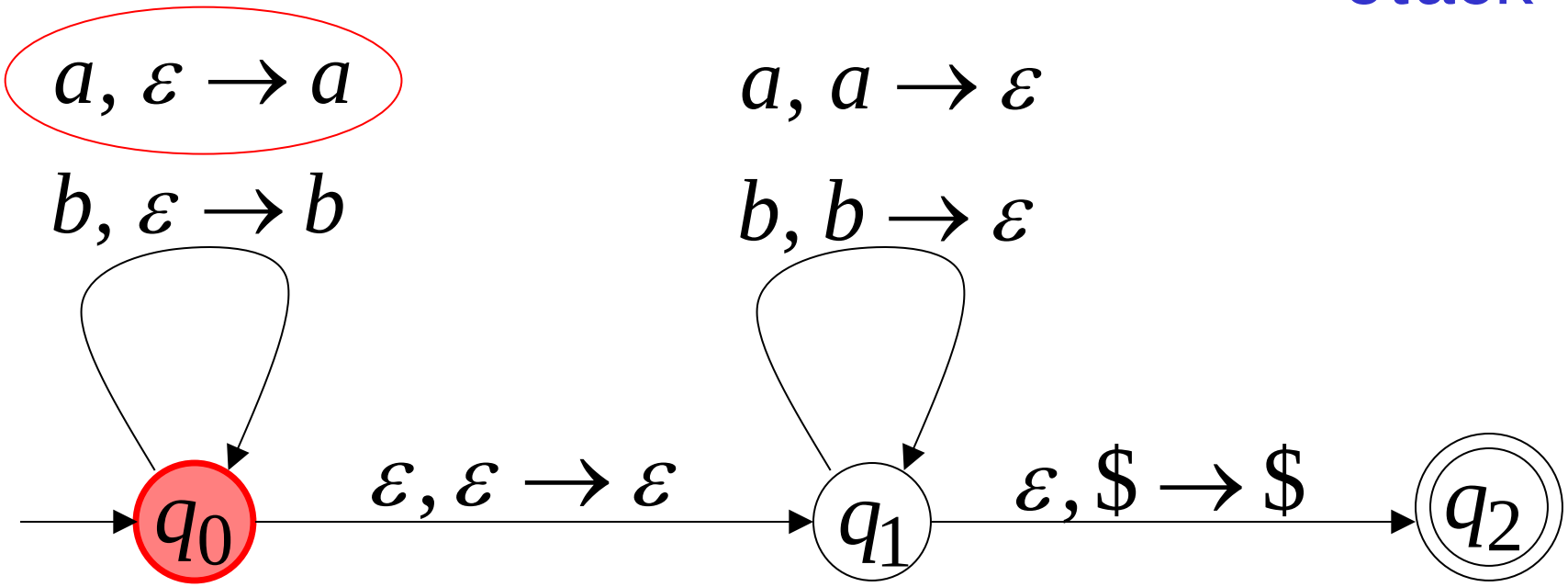
Time 1

Input



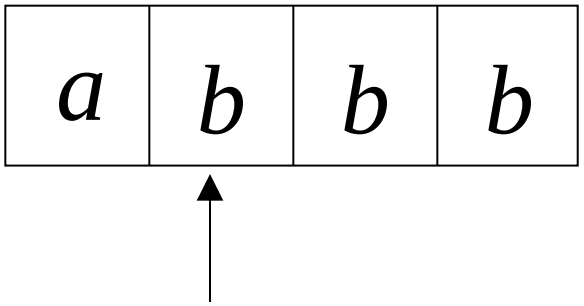


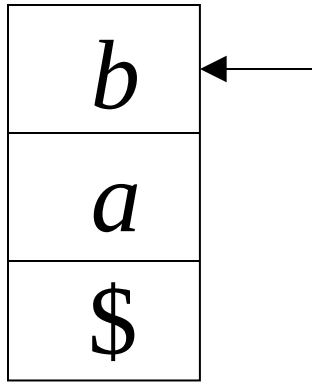
Stack



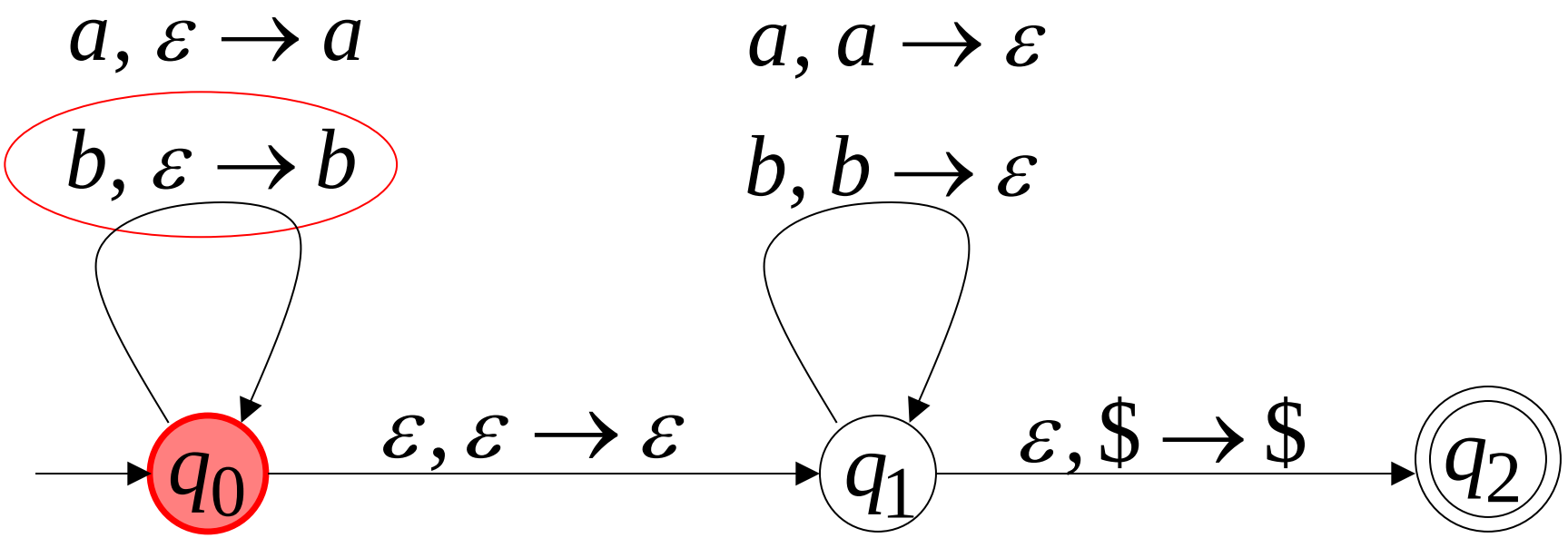
Time 2

Input



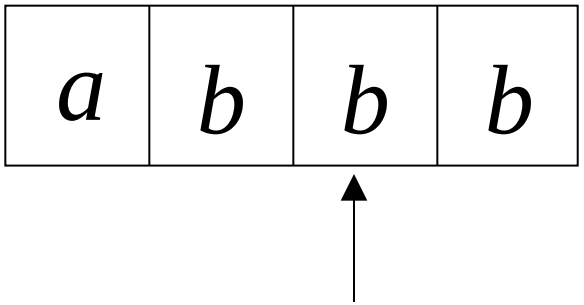


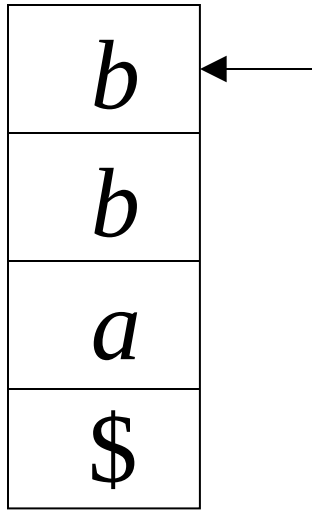
Stack



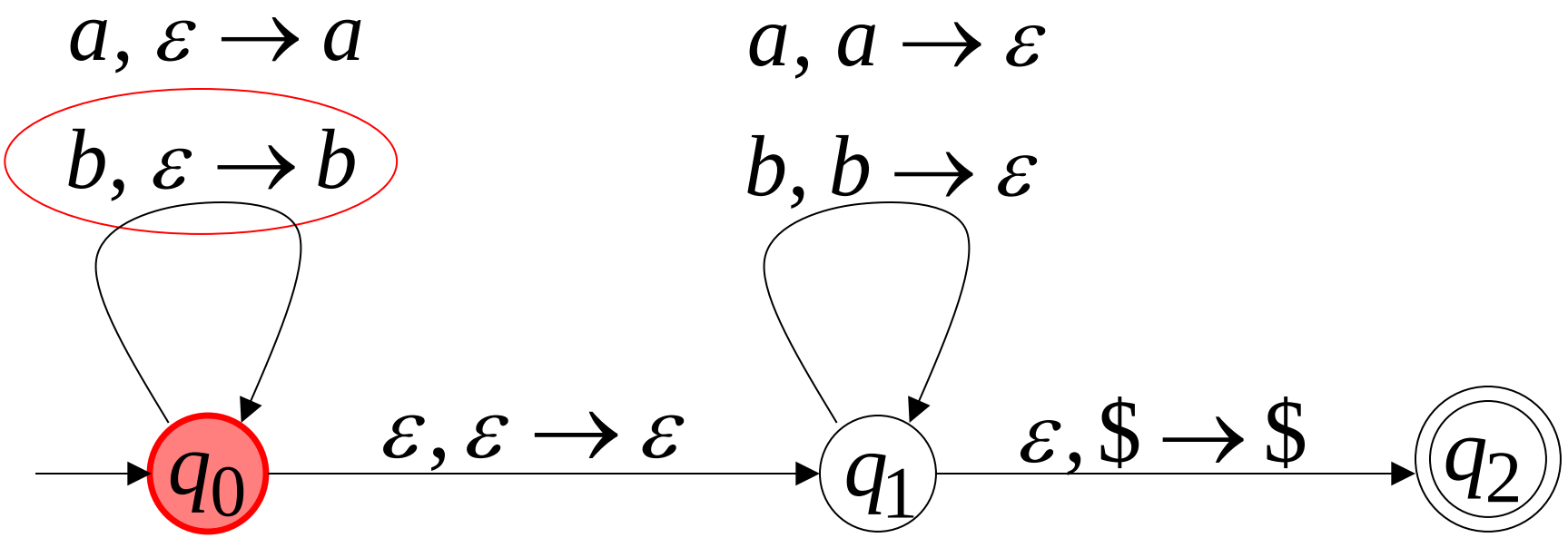
Time 3

Input





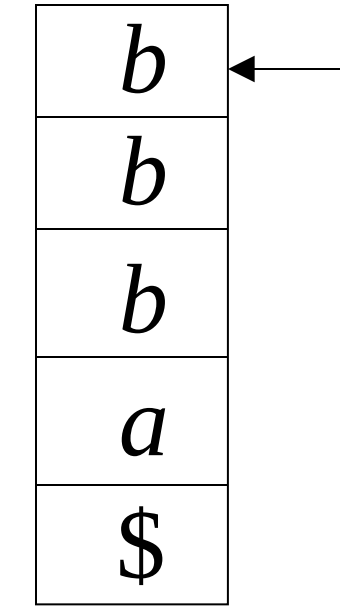
Stack



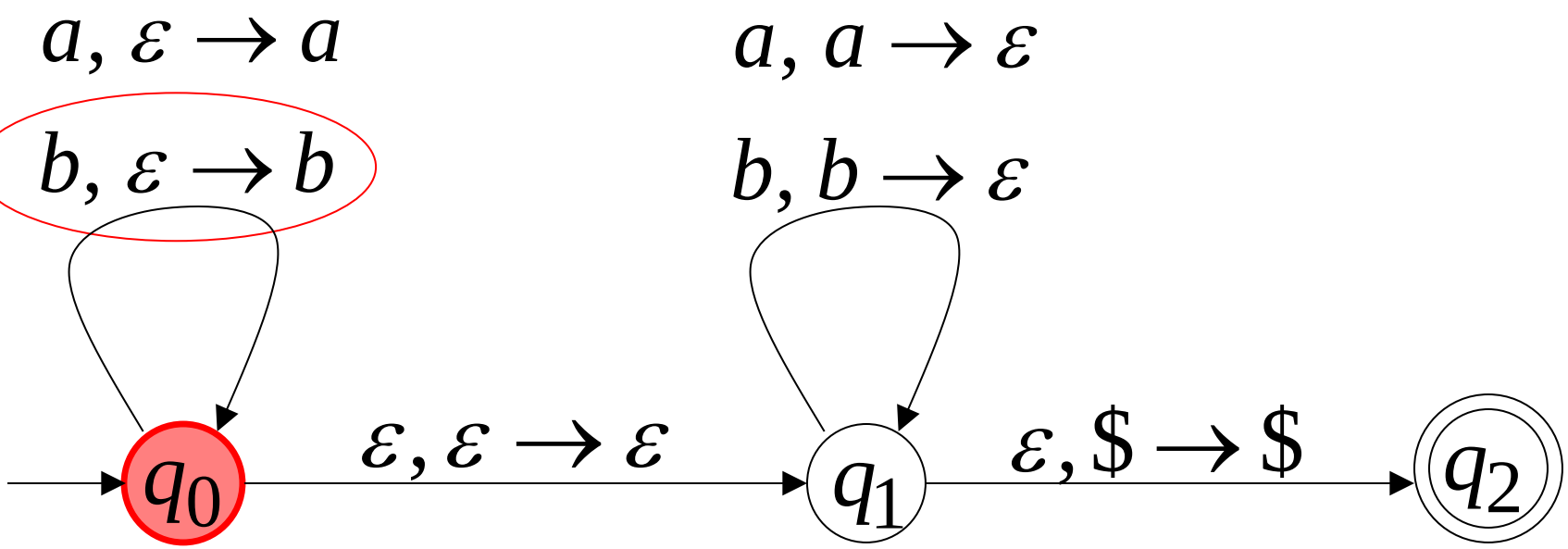
Time 4

Input

a	b	b	b



Stack



Time 5

Input

a	b	b	b

No accept state is reached

b
b
b
a
$\$$

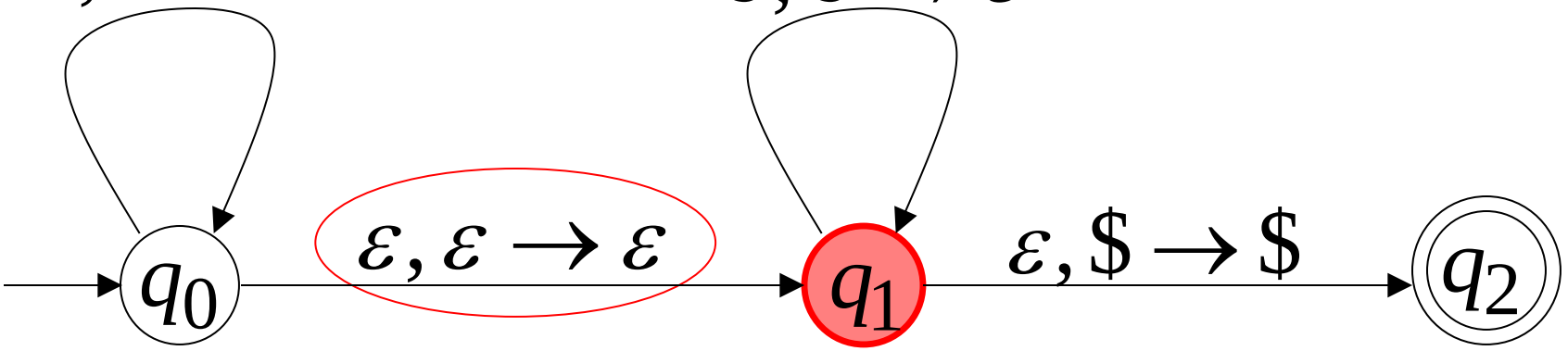
Stack

$a, \varepsilon \to a$

$b, \varepsilon \to b$

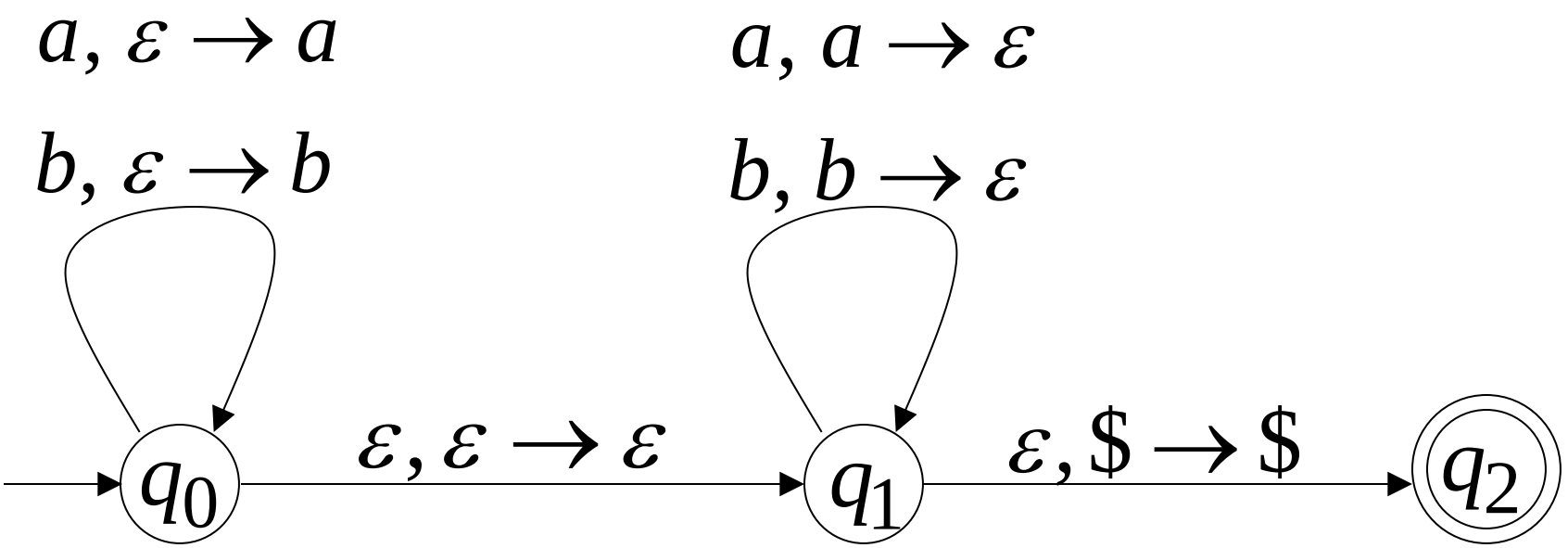
$a, a \to \varepsilon$

$b, b \to \varepsilon$

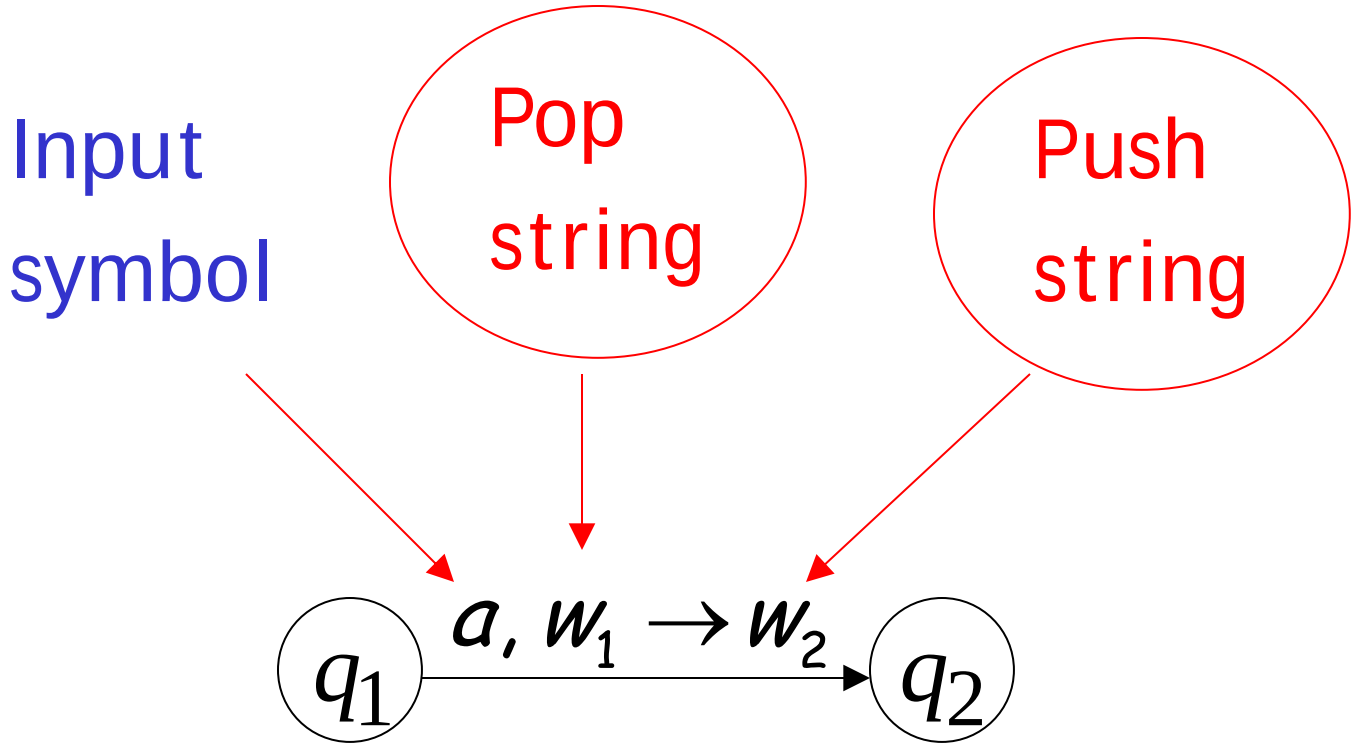


There is no computation that accepts string $abbb$

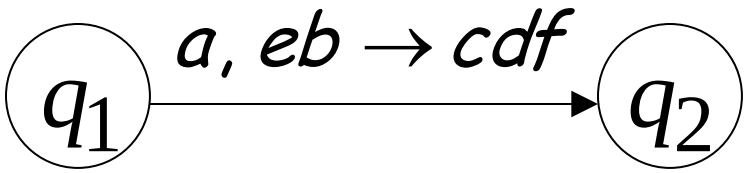
$$abbb \notin L(M)$$

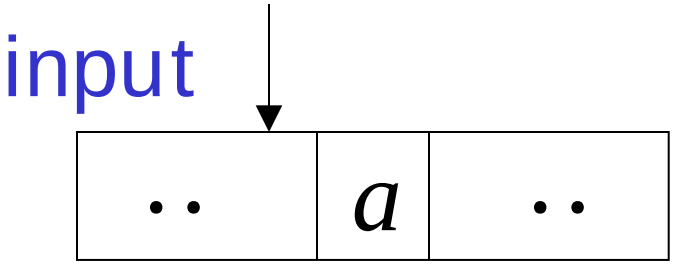


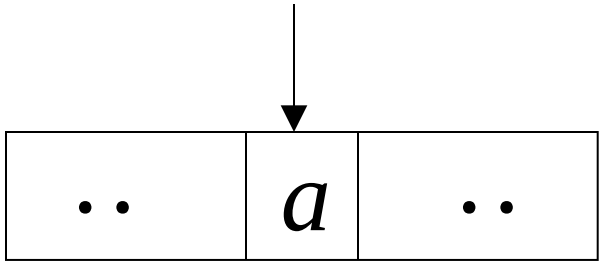
Pushing & Popping Strings

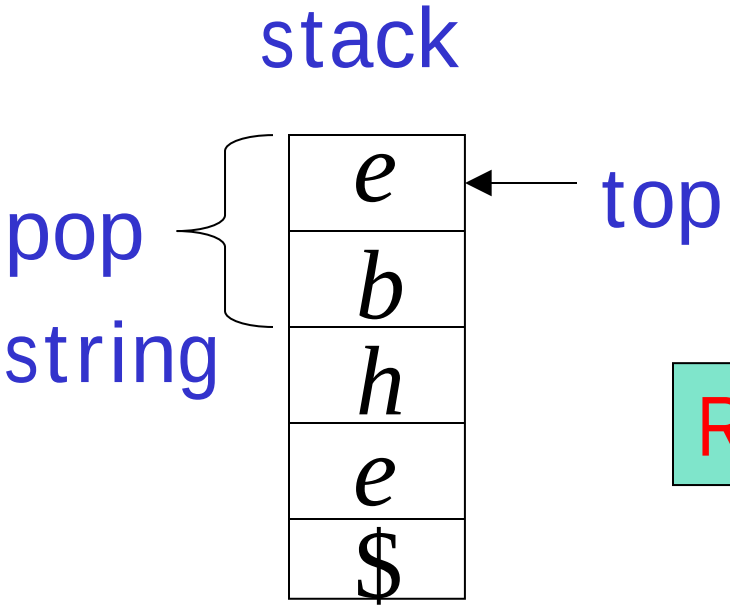


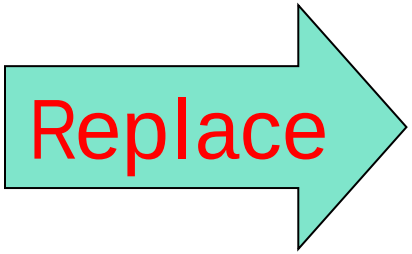
Example:

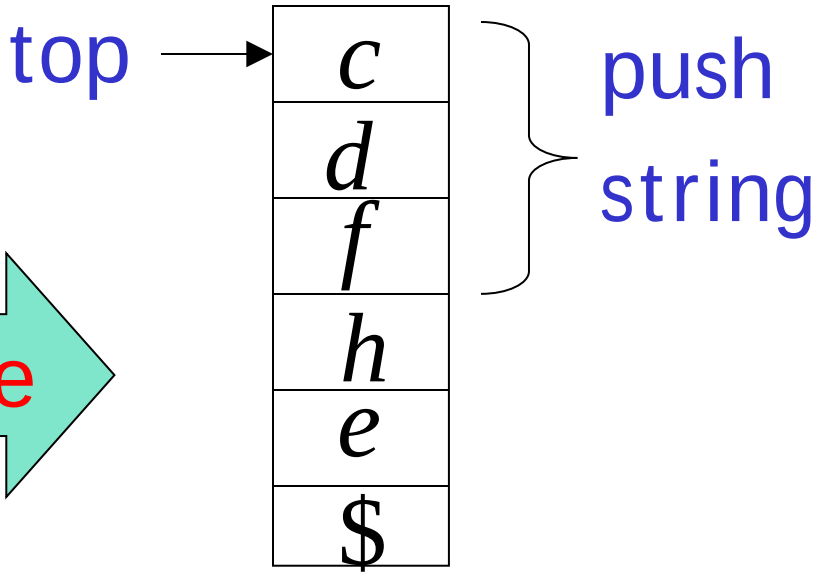

q_1
$a, eb \rightarrow cdf$
q_2

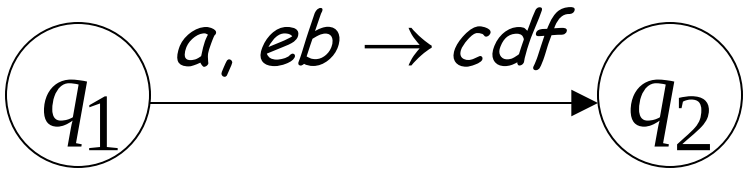

input
..
a
..

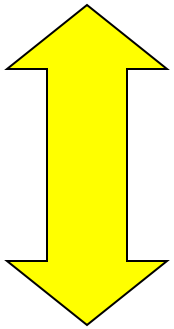

..
a
..


stack
pop
string
e
b
h
e
$
top


Replace

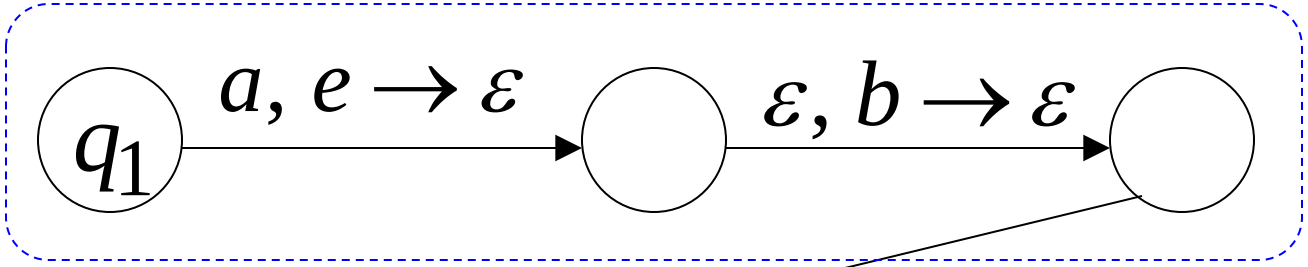

top
c
d
f
h
e
$
push
string

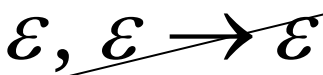




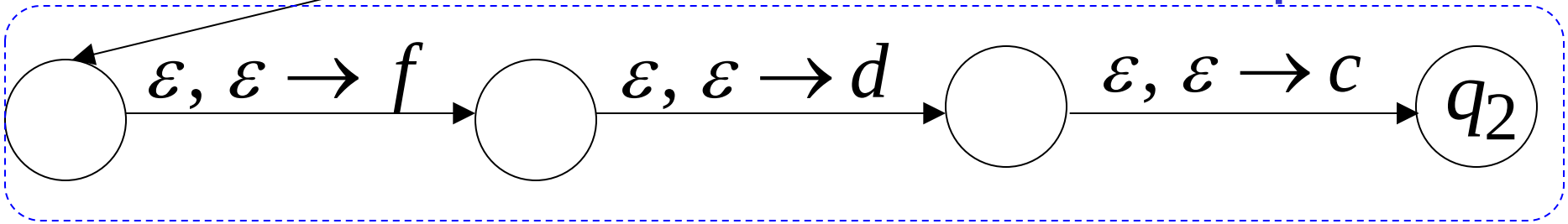
Equivalent transitions

pop





push



Another PDA example

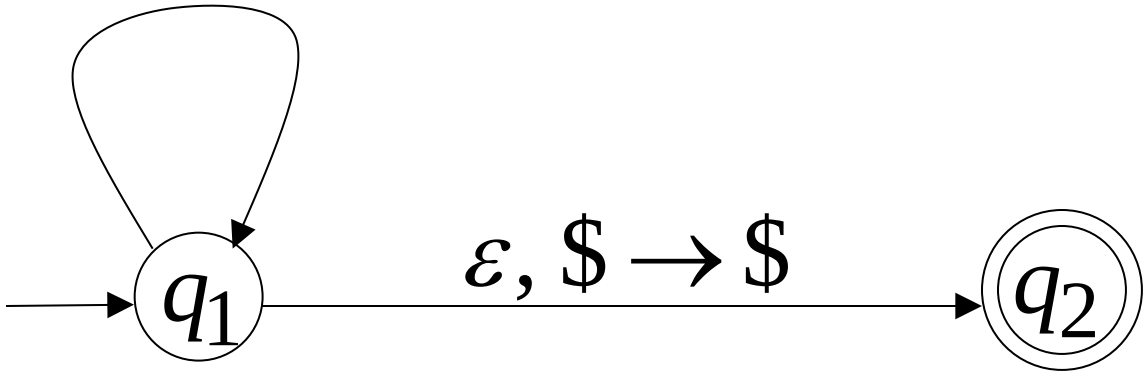
$$L(M) = \{w \in \{a,b\}^* : \quad n_a(w) = n_b(w)\}$$

PDA M

$a, \$ \to 0\$$ $\quad b, \$ \to 1\$$

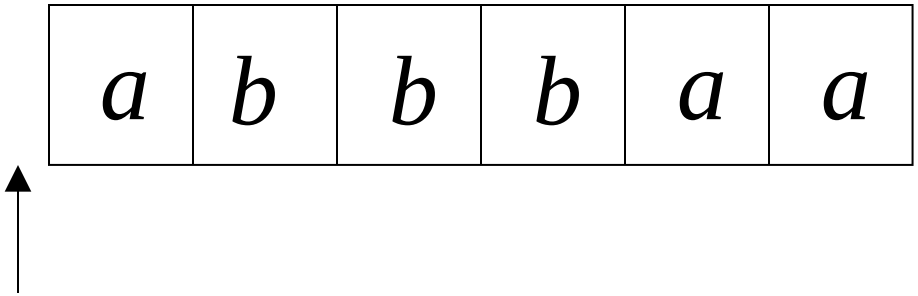
$a, 0 \to 00$ $\quad b, 1 \to 11$

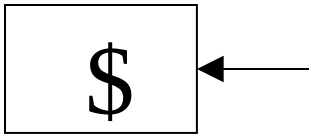
$a, 1 \to \varepsilon$ $\quad b, 0 \to \varepsilon$



Execution Example: Time 0

Input





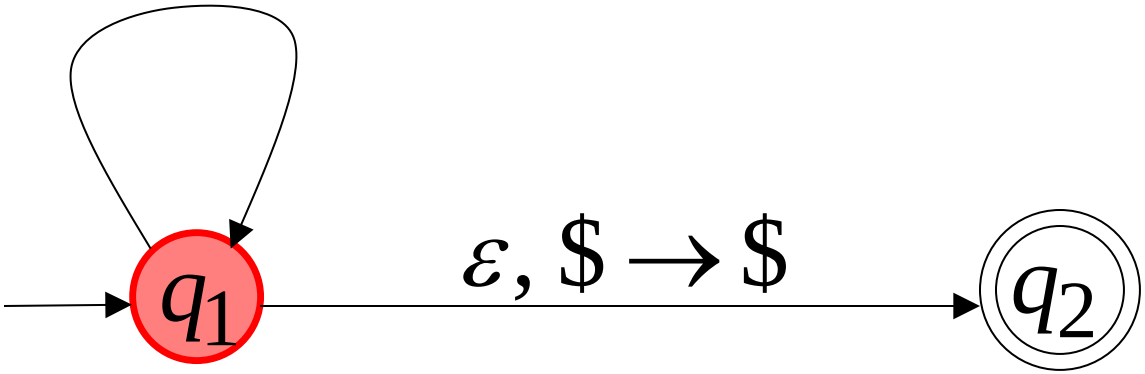
Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

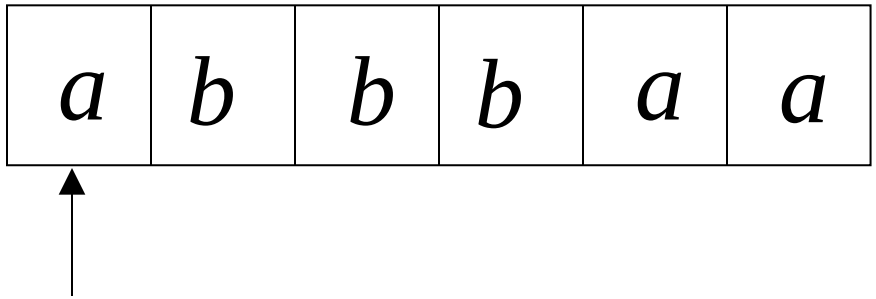
$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$

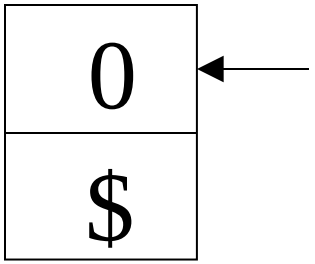
current state



Time 1

Input



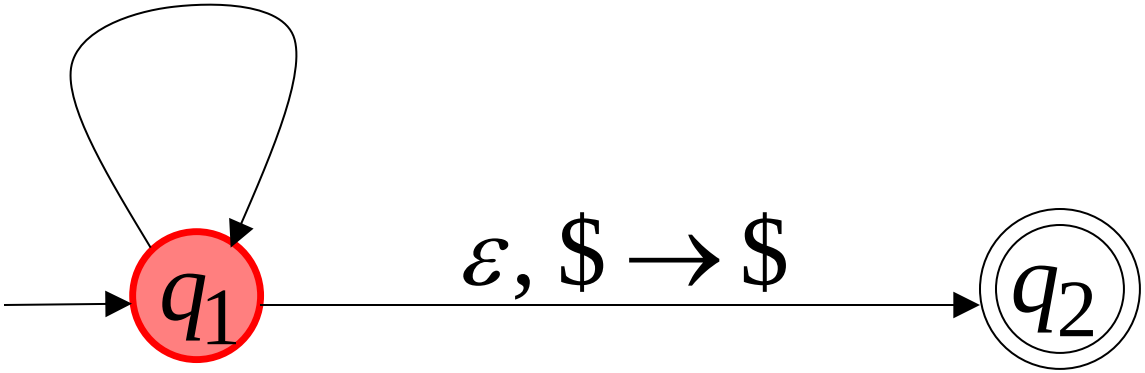


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

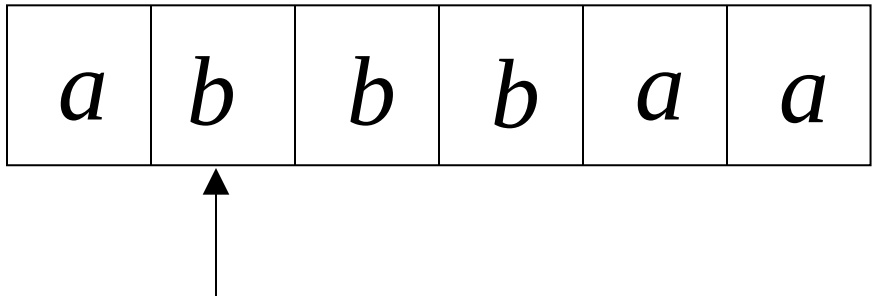
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

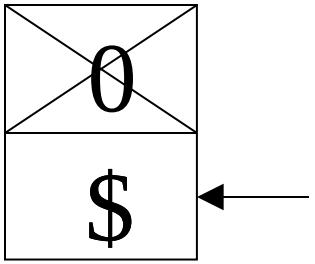
$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 3

Input



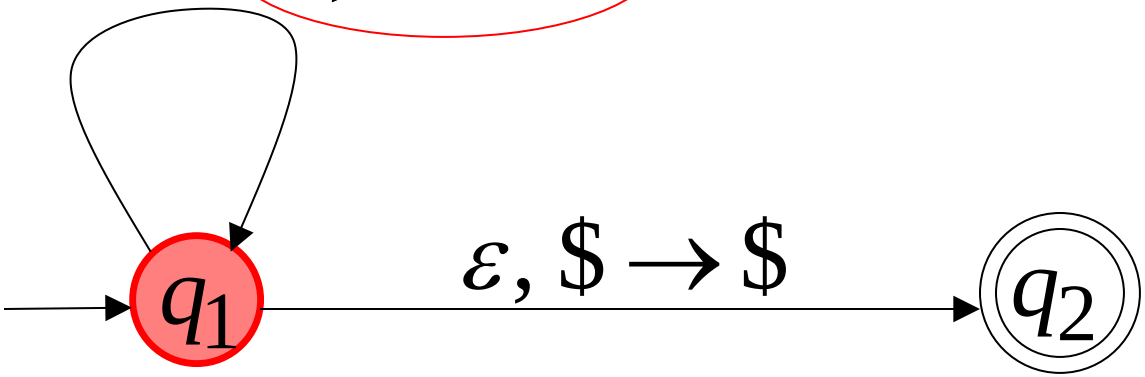


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

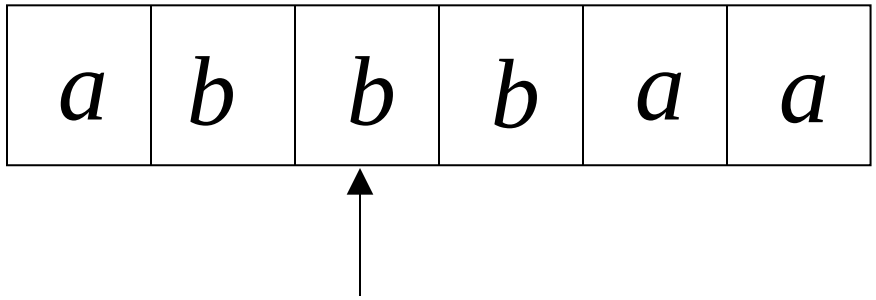
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

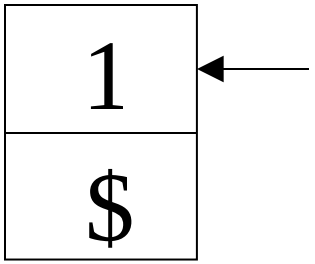
$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 4

Input



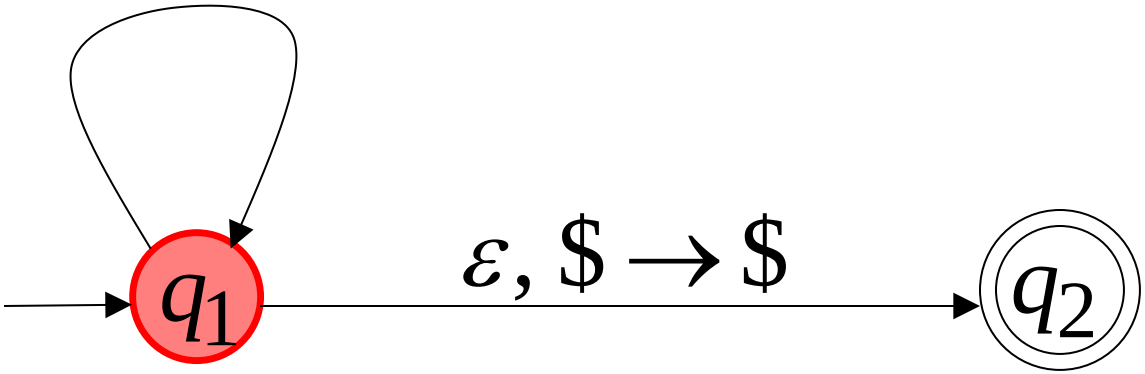


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

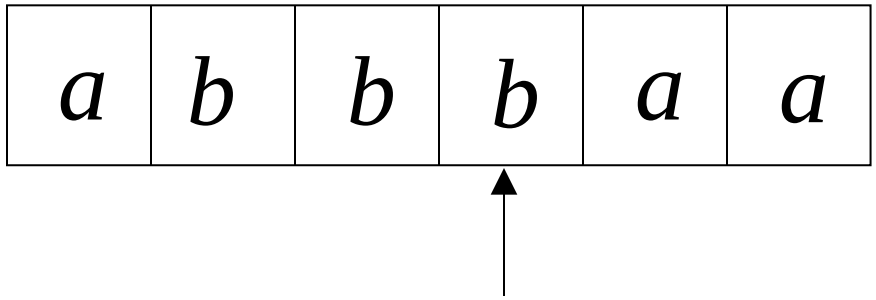
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

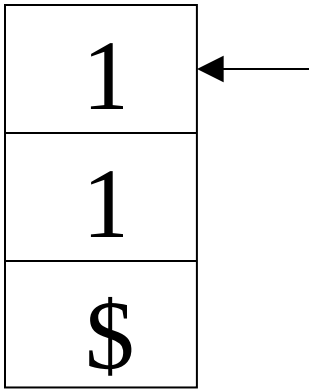
$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 5

Input



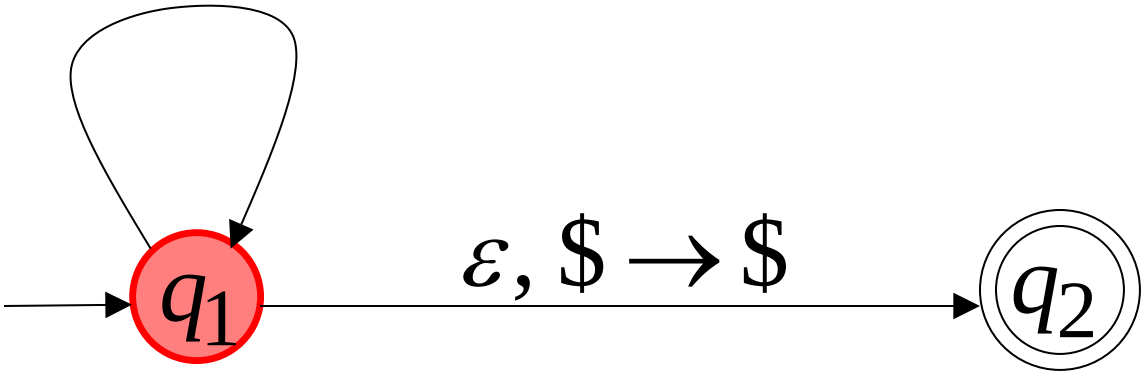


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

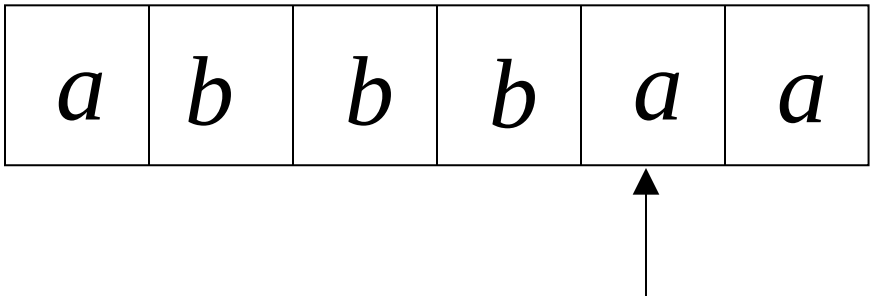
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

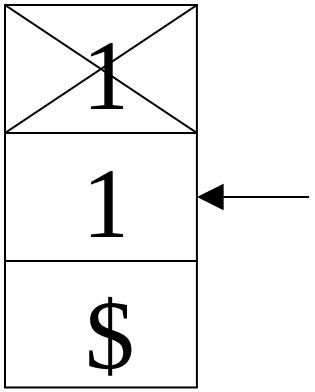
$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 6

Input



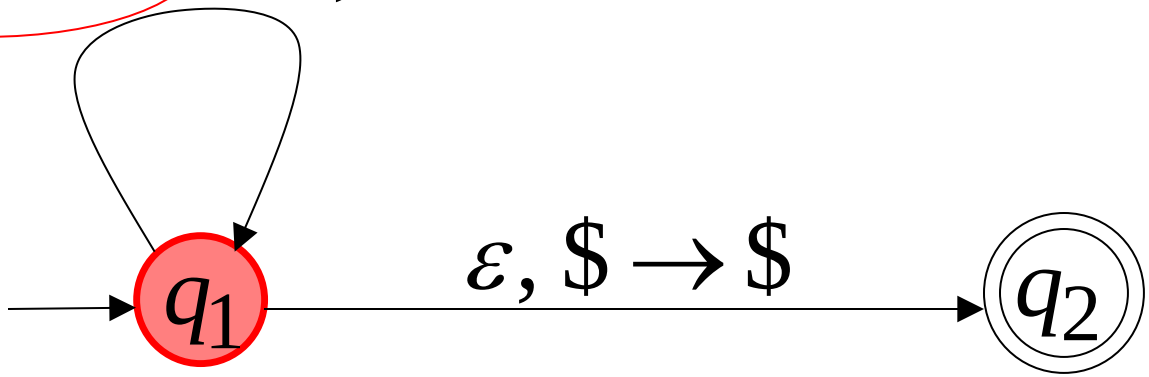


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

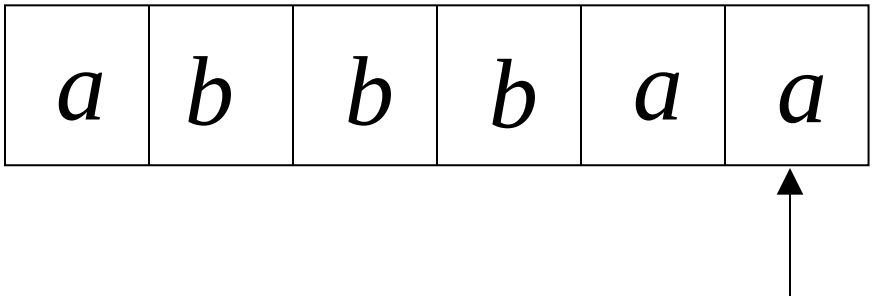
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

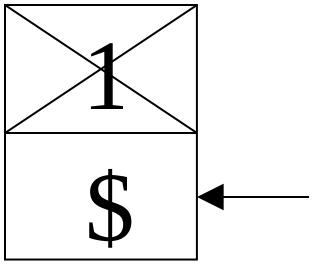
$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 7

Input



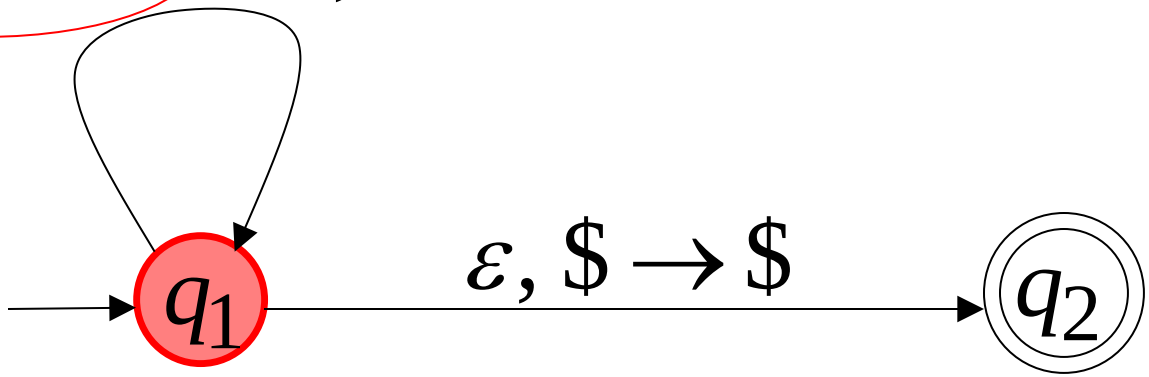


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

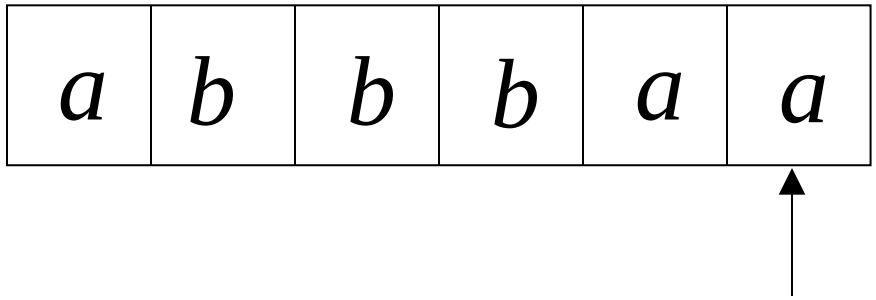
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

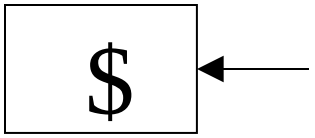
$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 8

Input



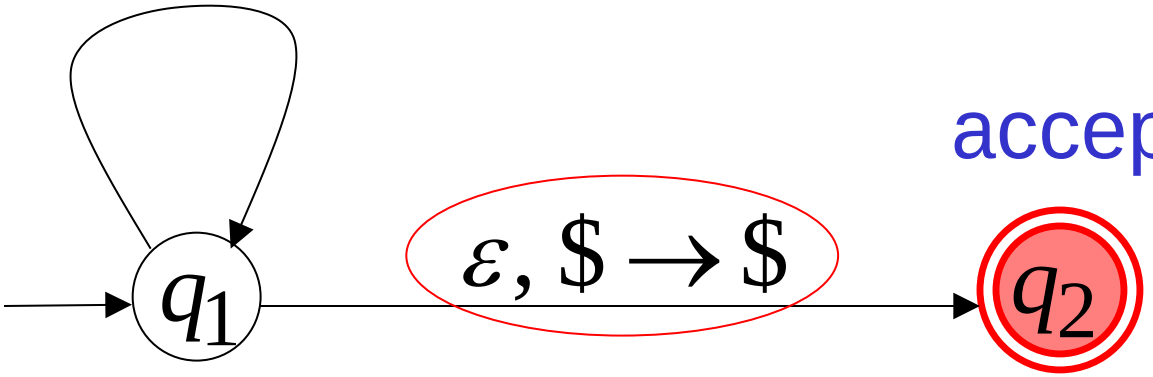


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

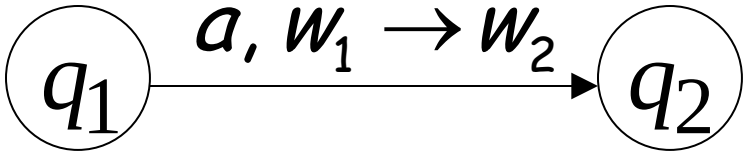
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



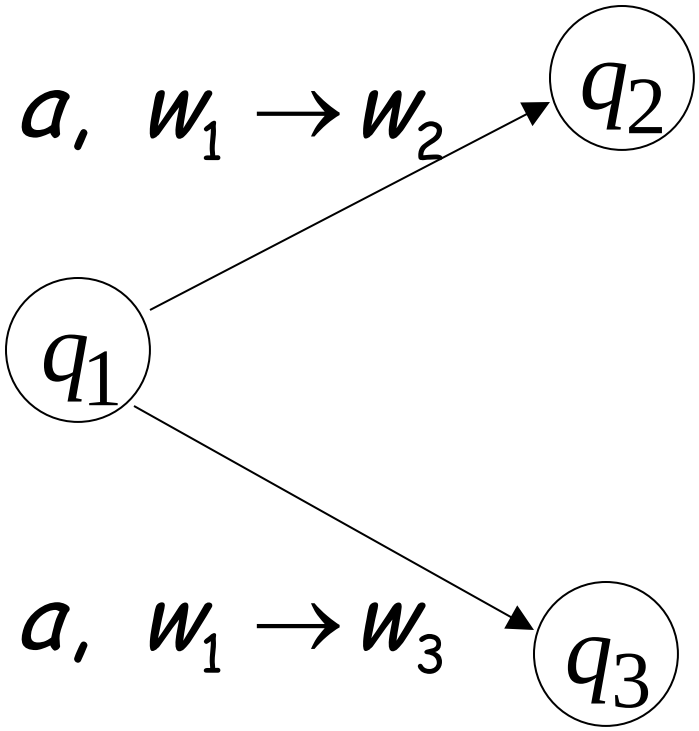
accept

Formalities for PDAs



Transition function:

$$\delta(q_1, a, w_1) = \{(q_2, w_2)\}$$



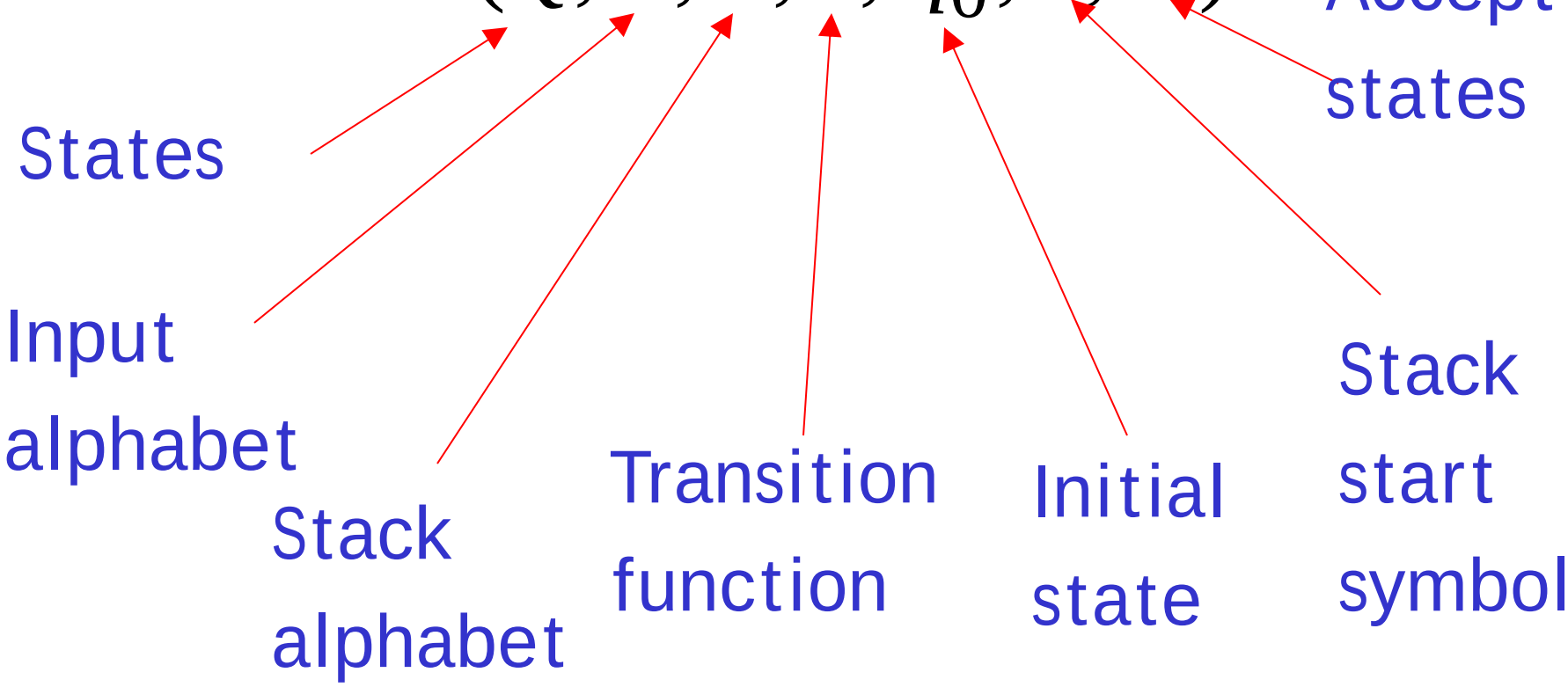
Transition function:

$$\delta(q_1, a, w_1) = \{(q_2, w_2),\ (q_3, w_3)\}$$

Formal Definition

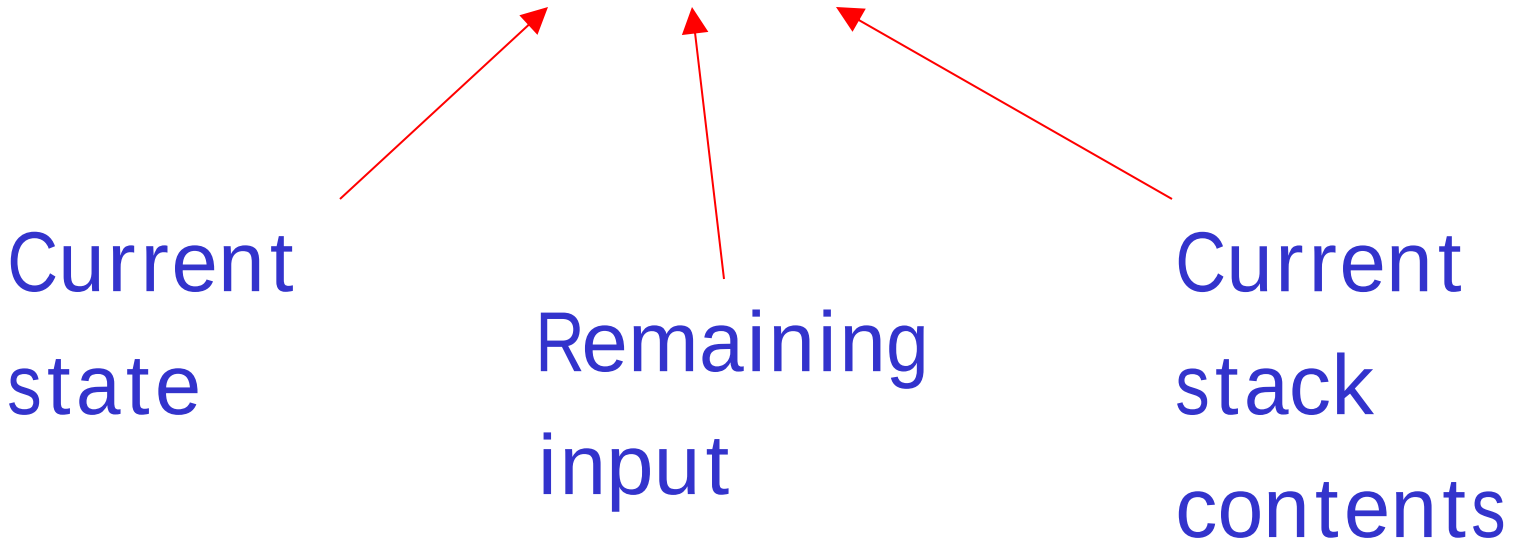
Pushdown Automaton (PDA)

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$$



Instantaneous Description

$$(q,\ u,\ s)$$

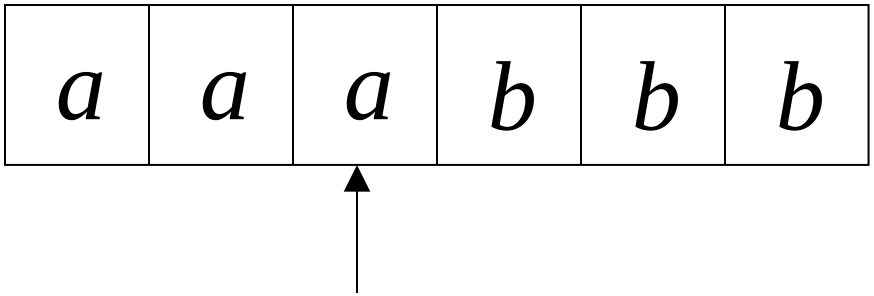


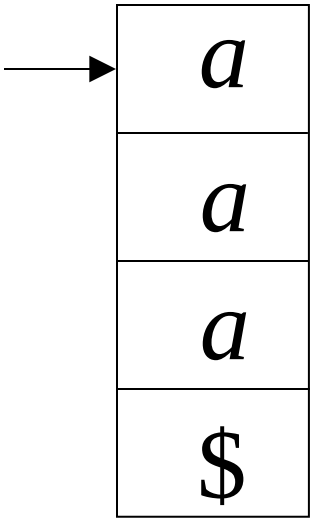
Example: Instantaneous Description

$$(q_1, bbb, aaa\$)$$

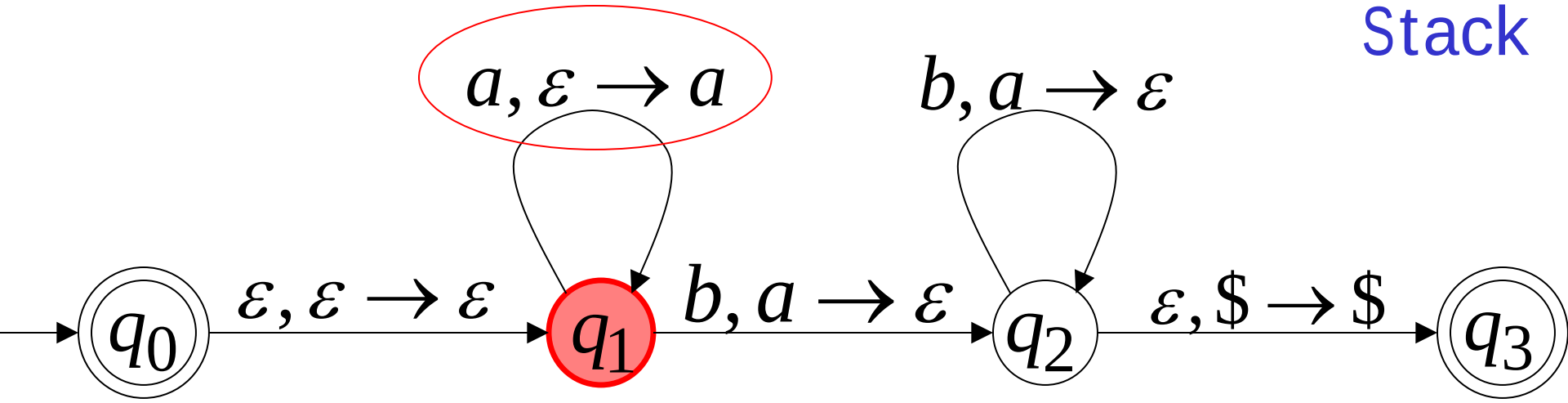
Time 4:

Input





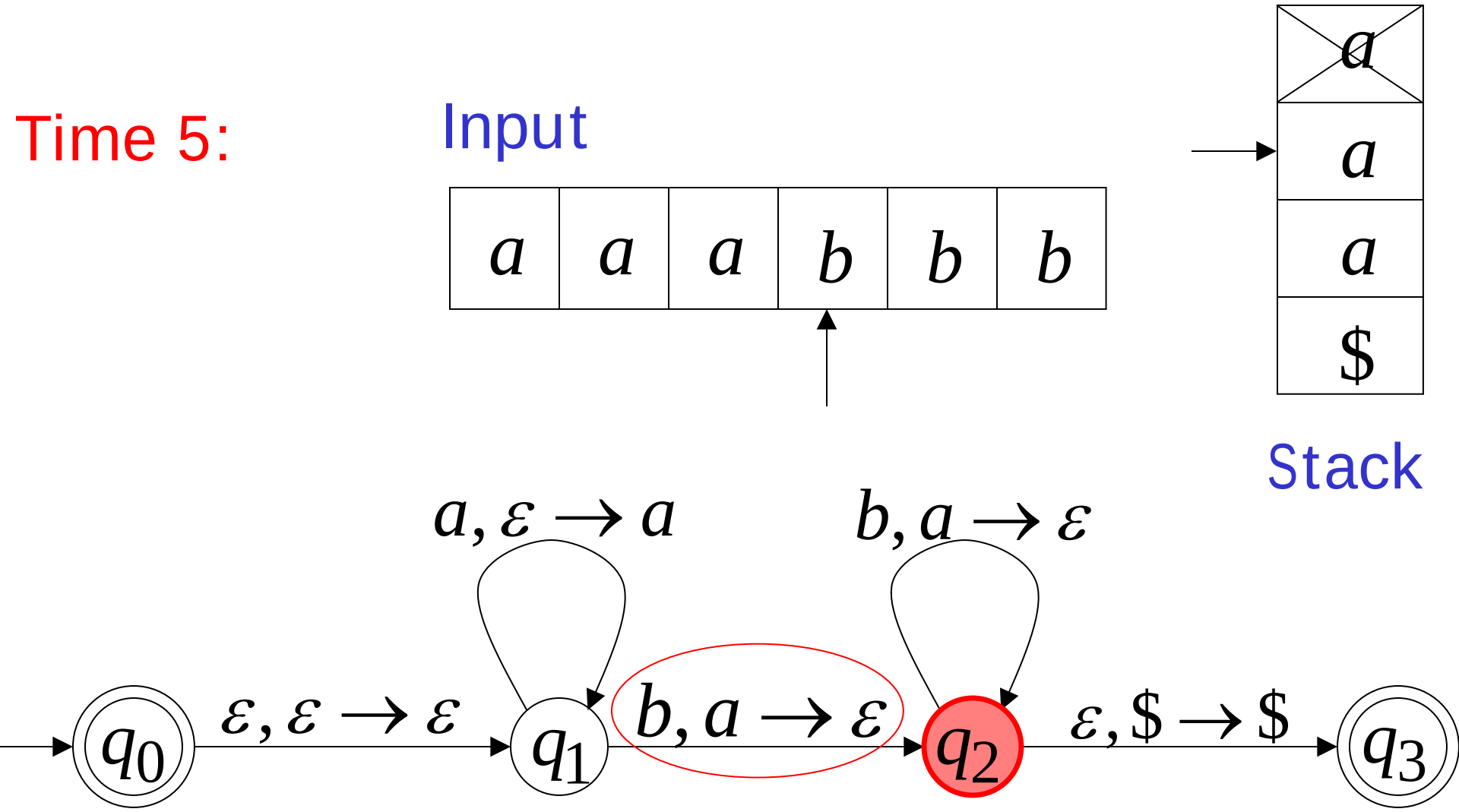
Stack



Example: Instantaneous Description

$$(q_2, bb, aa\$)$$

Time 5:



We write:

$$(q_1, bbb, aaa\$) \succ (q_2, bb, aa\$)$$

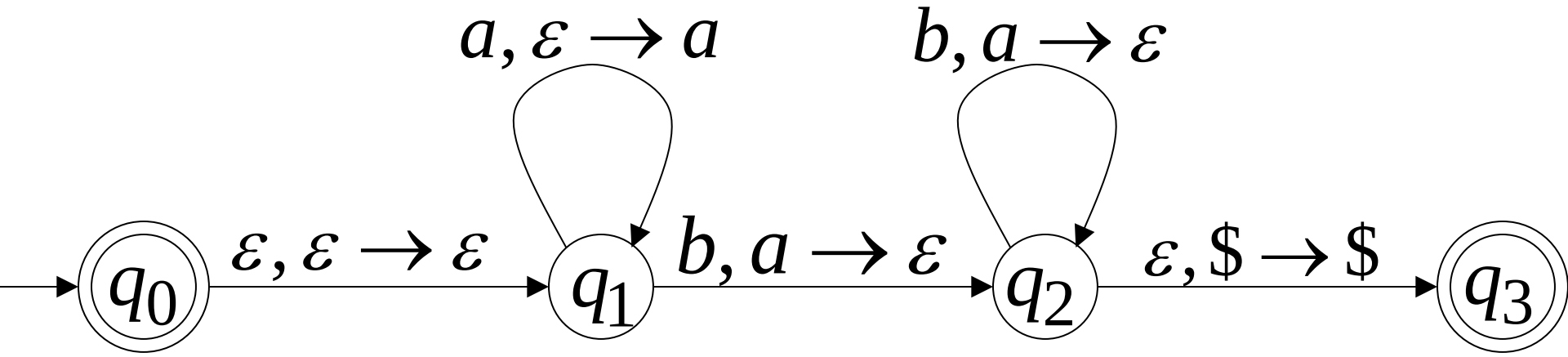
Time 4 Time 5

A computation:

$$(q_0, aaabbb, \$) \succ (q_1, aaabbb, \$) \succ$$

$$(q_1, aabbb, a\$) \succ (q_1, abbb, aa\$) \succ (q_1, bbb, aaa\$) \succ$$

$$(q_2, bb, aa\$) \succ (q_2, b, a\$) \succ (q_2, \varepsilon, \$) \succ (q_3, \varepsilon, \$)$$



$$(q_0, aaabbb, \$) \succ (q_1, aaabbb, \$) \succ$$
$$(q_1, aabbb, a\$) \succ (q_1, abbb, aa\$) \succ (q_1, bbb, aaa\$) \succ$$
$$(q_2, bb, aa\$) \succ (q_2, b, a\$) \succ (q_2, \varepsilon, \$) \succ (q_3, \varepsilon, \$)$$

For convenience we write:

$$(q_0, aaabbb, \$) \overset{*}{\succ} (q_3, \varepsilon, \$)$$

Language of PDA

Language $L(M)$ accepted by PDA M:

$$L(M) = \{w: \ (q_0, w, z) \overset{*}{\succ} (q_f, \varepsilon, s)\}$$

Initial state

Accept state

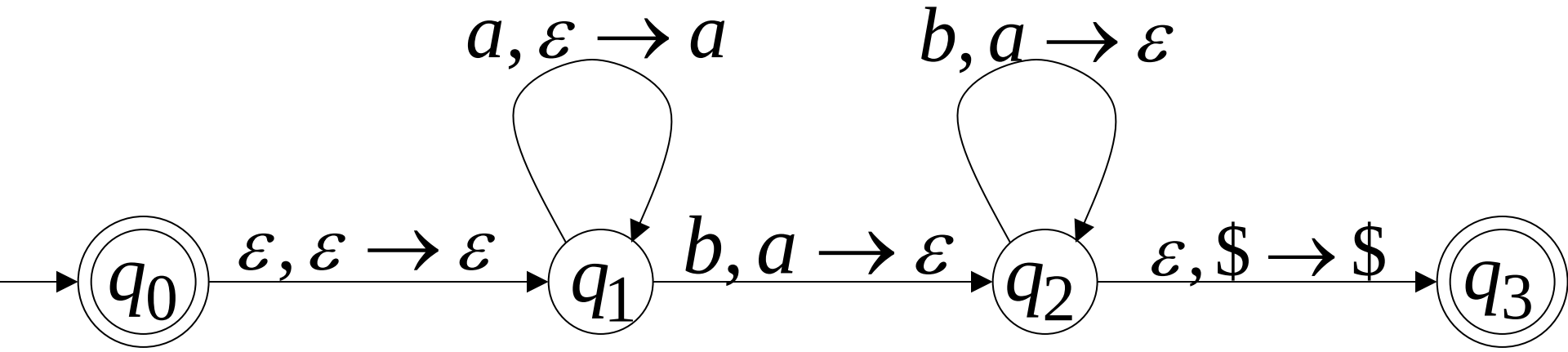
Example:

$$(q_0, aaabbb, \$) \overset{*}{\succ} (q_3, \varepsilon, \$)$$



$$aaabbb \in L(M)$$

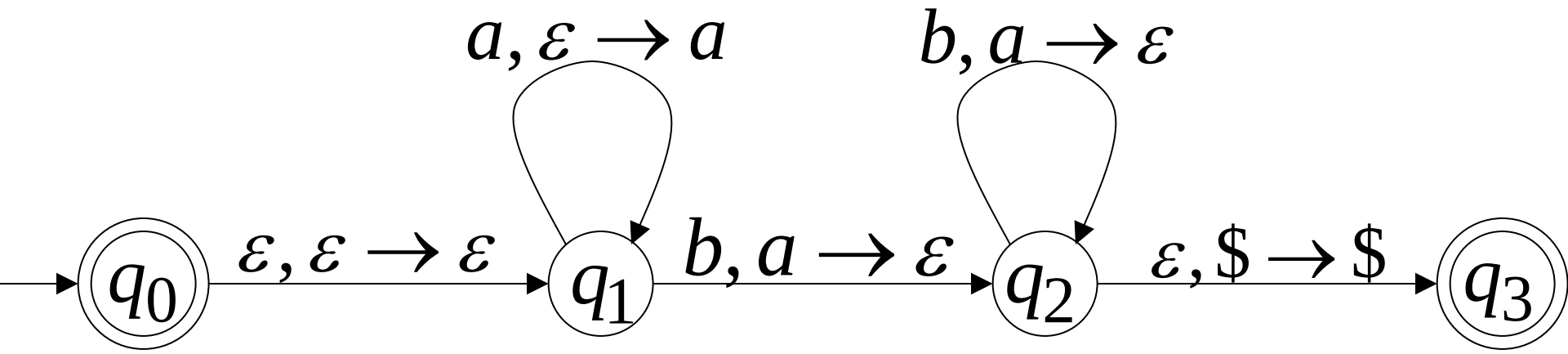
PDA M :



$$(q_0, a^n b^n, \$) \overset{*}{\succ} (q_3, \varepsilon, \$)$$

$$a^n b^n \in L(M)$$

PDA M :



Therefore: $L(M) = \{a^n b^n : n \geq 0\}$

PDA M :

